

# Asymptotic expansions of inverse functions on a real branch

Power-logarithmic reversion with real exponents, from two Mathematica Stack Exchange questions to a unified theory and a Wolfram Language package

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## Abstract

The inverse of  $f(x) = x + x^2(1 + \log x)$  on the positive real axis has no Taylor expansion at 0: its expansion is  $y - y^2(1 + \log y) + y^3(2 \log^2 y + 5 \log y + 3) - \dots$ , with a polynomial in  $\log y$  at every power. The inverse of  $x + x^{\sqrt{2}}$  has irrational exponents,  $y - y^{\sqrt{2}} + \sqrt{2} y^{2\sqrt{2}-1} - \dots$ . The built-in functions `InverseSeries`, `Series`, `Asymptotic` and `InverseFunction` of the Wolfram Language either fail on such inputs or return the germ at a complex root. The same algebra of finite *power-log expansions* also governs forward asymptotics with irrational exponents, such as  $(1 + x + x^{\sqrt{2}})^{\sqrt{2}}$  at infinity, where `Asymptotic` returns its input unchanged.

We develop an ordered calculus of power-log blocks with exact truncated arithmetic, marker Lagrange-Bürmann inversion, and an Euler-operator formula for every block of  $g(y)^r$ . The positive real branch is selected explicitly. An implicit-function argument in finitely many small parameters proves convergence, a Rouché argument supplies a conditional numerical majorant, and exponent cutoffs have remainders governed by the first omitted block. Forward errors and residuals are transported with the derivative scaling appropriate to the observable, including powers of inverses at infinity. A finite-smoothness theorem also covers perturbations outside the analytic power-log class. A logarithmic coordinate extends the method to Lambert cores, including the real inverse of  $x \log x$  near 0 and of  $x e^x$  at infinity, with a convergent expansion and explicit remainder scale. The Wolfram Language package `AsymptoticInverse` supplies exact-exponent forward and inverse engines and inert remainder descriptors. We distinguish implemented computations, hypotheses carried by the input data, and extensions that need larger scales, and compare the nine reports underlying the original implementation.

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\*Written with Claude (Anthropic). The theory synthesises nine independently written reports, listed in Section 14, and adds the extensions needed for the unified package.

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## 1 Introduction

### 1.1 Two questions

The first question [1] asks for an asymptotic expansion, as  $y \rightarrow 0^+$ , of the inverse of

$$f_1(x) = x + x^2(1 + \log x), \quad x > 0, \quad (1.1)$$

on the branch that maps  $(0, \infty)$  to itself. The natural attempts fail in instructive ways. `AsymptoticSolve[x + x^2 (1 + Log[x]) == y, {x}, y -> 0]` returns unevaluated; `InverseSeries` returns unevaluated because the forward series has a coefficient containing  $\log x$ ; and

```
Series[InverseFunction[Function[x, x + x^2 (1 + Log[x])]]][y], {y, 0, 2}]
```

returns the perfectly correct Taylor germ

$$c + \frac{y}{c-1} - \frac{3+2W_{-1}(-e)}{2(c-1)^3} y^2 + \mathcal{O}(y^3), \quad c = e^{W_{-1}(-e)-1} \approx -0.1178 - 0.5332i, \quad (1.2)$$

of the inverse at a *different* preimage of 0: the nonzero complex root  $c$  of  $1 + x(1 + \log x) = 0$ , where  $f_1$  is analytic and  $f_1'(c) = c - 1 \neq 0$  (Section 4). The real branch has no Taylor expansion at all. Its expansion is

$$\begin{aligned} g_1(y) = & y - y^2(1 + L) + y^3(2L^2 + 5L + 3) - y^4\left(5L^3 + \frac{41}{2}L^2 + 27L + \frac{23}{2}\right) \\ & + y^5\left(14L^4 + \frac{241}{3}L^3 + \frac{335}{2}L^2 + 151L + \frac{299}{6}\right) + \mathcal{O}(y^6(1 + |L|)^5), \end{aligned} \quad (1.3)$$

with  $L = \log y$ : every power of  $y$  carries a polynomial in  $\log y$ , of degree growing with the power, so the scale of the expansion is neither a Taylor nor a Puiseux scale. The question also asks, as a second example, for

$$g_2(y) = y - y^{\sqrt{2}} + \sqrt{2}y^{2\sqrt{2}-1} - \frac{6-\sqrt{2}}{2}y^{3\sqrt{2}-2} + \left(\frac{17\sqrt{2}}{3} - 4\right)y^{4\sqrt{2}-3} + \mathcal{O}(y^{5\sqrt{2}-4}), \quad (1.4)$$

the inverse of  $f_2(x) = x + x^{\sqrt{2}}$ : no logarithms this time, but exponents in the semigroup  $1 + (\sqrt{2} - 1)\mathbb{N}$ , which no rational grid contains.

The second question [2] asks for the forward asymptotics of  $(1 + x + x^{\sqrt{2}})^{\sqrt{2}}$  as  $x \rightarrow \infty$ . The built-in `Asymptotic` correctly returns its leading term  $x^2$ , but adding `SeriesTermGoal -> 7` returns the input unchanged. The expansion is

$$\begin{aligned} (1 + x + x^{\sqrt{2}})^{\sqrt{2}} = & x^2 + \sqrt{2}x^{3-\sqrt{2}} + \left(1 - \frac{1}{\sqrt{2}}\right)x^{4-2\sqrt{2}} + \left(\frac{2\sqrt{2}}{3} - 1\right)x^{5-3\sqrt{2}} \\ & + \sqrt{2}x^{2-\sqrt{2}} + \frac{13-9\sqrt{2}}{12}x^{6-4\sqrt{2}} + (2 - \sqrt{2})x^{3-2\sqrt{2}} + \mathcal{O}(x^{7-5\sqrt{2}}), \end{aligned} \quad (1.5)$$

and its inverse, as  $y \rightarrow \infty$ , begins  $\sqrt{y} - \frac{1}{\sqrt{2}}y^{(2-\sqrt{2})/2} + \frac{3-\sqrt{2}}{4}y^{(3-2\sqrt{2})/2} + \mathcal{O}(y^{(4-3\sqrt{2})/2})$ .

## 1.2 What is common to all of this

All these expansions are finite or infinite sums of *power-log blocks*  $w^\beta P(\log w)$ , where  $w \rightarrow 0^+$  is a local variable ( $w = y$ ,  $w = 1/x$ ,  $\dots$ ),  $\beta$  is a real exponent, and  $P$  is a polynomial. Their supports lie in translates of semigroups generated by finitely many positive gaps. These semigroups are locally finite, even when the groups generated by the same gaps are dense in  $\mathbb{R}$ , so every truncation is a finite computation. The logarithmic polynomials are handled by the Euler identity  $\frac{d}{dw}[w^\beta P(\log w)] = w^{\beta-1}(\beta + \frac{d}{dL})P(L)$ ; and remainders must keep their logarithmic factor, because  $y^3(2\log^2 y + 5\log y + 3)$  is not  $\mathcal{O}(y^3)$ .

This article gives the complete theory of this algebra as far as it is needed to compute and to justify such expansions, forward and inverse, with proofs, and describes the package `AsymptoticInverse` that implements it. It grew out of nine independently written reports on the first question (Section 14 and [12]), which agree on the core of the theory and each contribute something the others lack; here their contents are unified, the proofs are completed, and the theory is extended to finite endpoints from either side, to infinity, to powers and logarithms of the inverse, and to the forward expansion problem of the second question.

## 1.3 Summary of results

- **Scale** (Section 2). Power-log blocks are totally ordered by size, the finitely generated exponent semigroups are locally finite, and the truncated arithmetic (products, real powers, logarithms, exponentials, composition with polynomials in the logarithm) is exact and terminates. The single primitive is the composition recurrence  $Q_{k+1} = ((a - k)Q_k + Q'_k)/(k + 1)$  for  $(1 + t)^a P(L + \log(1 + t))$ .
- **Forward expansions** (Section 3). Expressions built from constants, the variable, arithmetic, real powers, logarithms, bounded exponentials and analytic functions of expressions with a limit have convergent power-log expansions; a precision-tracking evaluator produces them with a rigorous remainder class, which answers the second question.
- **Branch** (Section 4). For  $f(x) = ax^p(1 + \sum_i x^{\delta_i} B_i(\log x))$  with  $a, p \neq 0$ ,  $\delta_i > 0$ , the real branch is the one with  $x > 0$  and  $x/(y/a)^{1/p} \rightarrow 1$ ; it is unique, and the complex germ (1.2) is explained.
- **Lagrange-Bürmann with a marker** (Section 5). A residue proof of  $\Phi(u) = \Phi(q) + \sum_N \frac{(-\eta)^N}{N!} \partial_q^{N-1} [\Phi'(q) h(q)^N]$  at a positive point, valid analytically and formally, with the transport of observables and the general-core form.
- **Coefficients** (Section 6). For every multi-index  $\mathbf{k}$  of weight  $\sigma$ , the block of  $g(y)^r$  at  $z^{r+\sigma}$ ,  $z = (y/a)^{1/p}$ , is  $\frac{(-1)^{|\mathbf{k}|} r}{p^{|\mathbf{k}|} \mathbf{k}!} \prod_{j=1}^{|\mathbf{k}|-1} (\mathcal{D} + r + \sigma + pj) \prod_i B_i(L)^{k_i}$ ,  $\mathcal{D} = d/dL$ ; equal weights are added. Degrees, leading logarithms (Catalan numbers), closed forms for constant coefficients (Fuss-Catalan),  $\log g$ , and the formal uniqueness through the triangular equation, Picard iteration and Newton iteration.
- **Convergence** (Section 7). The expansion converges absolutely for small positive  $y$  after substituting finitely many small parameters  $z^{\delta_i}(\log z)^j$ , with coefficient bounds  $|C_{\mathbf{k}}(L)| \leq MK^{|\mathbf{k}|}(1 + |L|)^{D(\mathbf{k})}$ ; an explicit conditional tail majorant follows from Rouché's theorem.
- **Remainders** (Section 8). Truncation below any exponent cutoff leaves  $z^{1+\sigma_*} C_{\sigma_*}(L) + o(z^{1+\sigma_*})$  where  $\sigma_*$  is the first omitted weight and  $C_{\sigma_*}$  the complete coefficient there, computed from the one-step boundary of the retained multi-index set; depth truncation has its own bound.

- **Transport** (Section 9). Forward remainders and residuals are transported to the inverse with the correct scaling  $z^{1-p}$ ; a residual bracket gives existence and a two-sided error bound; and a theorem with a complete proof shows that finite smoothness of the perturbation, with derivative bounds in the power–log scale, suffices for the marker formula with an explicit remainder, which covers perturbations such as  $x^2 \sin(\log x)$ .
- **Endpoints** (Section 10). Finite points from either side and  $\pm\infty$  reduce to the same engine through a local variable and a signed leading power  $p \neq 0$ ; the limit  $y_0$  may be finite or infinite; powers  $(x - x_0)^r$  and  $x^r$  of the inverse are obtained directly. Worked examples include the Wright omega function and the inverse of (1.5).
- **Lambert coordinates and boundaries** (Section 12). Leading logarithmic and exponential cores reduce to a convergent Lambert expansion in a reciprocal-logarithm coordinate, with a polynomial in its logarithm at every order. This covers  $x \log x$  near 0 and  $xe^x$  at infinity. Zero gaps, flat perturbations and oscillatory coefficients require further scale choices; symbolic exponents can use depth truncation when their branch assumptions are proved.
- **Package and reports** (Sections 13, 14). Data structures, exact comparison of algebraic exponents, the two engines, precision rules, checks; and the comparison of the nine reports, whose Wolfram packages were run natively for the first time.

## 1.4 Conventions

$\mathbb{N} = \{0, 1, 2, \dots\}$ . For a local variable  $w \rightarrow 0^+$  we write  $L = \log w$  and  $M(w) = 1 + |\log w|$ .  $\mathcal{O}$  and  $o$  refer to  $w \rightarrow 0^+$  with all other data fixed; no uniformity in parameters is asserted unless stated.  $\mathcal{D} = d/dL$  acts on polynomials in  $L$ . For a multi-index  $\mathbf{k} = (k_1, \dots, k_m) \in \mathbb{N}^m$  we write  $|\mathbf{k}| = \sum k_i$ ,  $\mathbf{k}! = \prod k_i!$ , and  $\mathbf{k} \cdot \boldsymbol{\delta} = \sum k_i \delta_i$ . Real powers of positive quantities are  $w^\beta = e^{\beta \log w}$ ; complex powers are never used on the real branch. Cutoffs are *exclusive*: a cutoff  $q$  retains the blocks of exponent  $< q$ . Wolfram Language names are set in typewriter font.

## 2 The power–log scale and its exact arithmetic

### 2.1 Blocks and their order

Throughout,  $w$  denotes a positive real variable tending to  $0^+$ ,  $L = \log w \rightarrow -\infty$  and  $M(w) = 1 + |\log w|$ .

**Definition 2.1** (Blocks). A *power–log block* is an expression  $w^\beta P(L)$  with  $\beta \in \mathbb{R}$  and  $P \in \mathbb{R}[L]$ ,  $P \neq 0$ . Its *exponent* is  $\beta$  and its *logarithmic degree* is  $\deg P$ .

**Lemma 2.2** (Power beats logarithm). *For every  $\varepsilon > 0$  and every real  $b$ ,  $w^\varepsilon M(w)^b \rightarrow 0$  as  $w \rightarrow 0^+$ . Consequently, if  $\beta < \gamma$  and  $P, Q \neq 0$ , then  $w^\gamma Q(L) = o(w^\beta P(L))$ ; and within a fixed exponent, a higher logarithmic degree dominates:  $w^\beta L^j = o(w^\beta L^k)$  for  $j < k$ .*

*Proof.* Put  $w = e^{-t}$ ,  $t \rightarrow +\infty$ :  $w^\varepsilon M(w)^b = e^{-\varepsilon t} (1+t)^b \rightarrow 0$ . For the second claim the ratio is  $w^{\gamma-\beta} Q(L)/P(L)$ , and  $|Q(L)/P(L)| \leq CM(w)^{\deg Q}$  for  $w$  small because  $P(L)$  has no zeros for  $L$  sufficiently negative and  $|P(L)| \geq c|L|^{\deg P} \geq c$  there; the first claim finishes. The third claim is clear.  $\square$

The growth classes of nonzero blocks are totally ordered by increasing exponent and, at equal exponents, by decreasing logarithmic degree; blocks of the same exponent and degree have the same growth class. A finite sum of blocks with distinct exponents is asymptotic to

its block of least exponent (whole polynomial included), and vanishes identically only if every block does. This is why truncations are taken by *complete blocks*: cutting a block inside its polynomial would drop terms larger than the retained terms of the next exponent.

**Definition 2.3** (Remainder classes). For  $\beta \in \mathbb{R}$  and an integer  $k \geq 0$  we write  $R(w) = \mathcal{O}_{\beta,k}$ , or in the package `PowerLogRemainder[w,  $\beta$ , k]`, for  $R(w) = \mathcal{O}(w^\beta M(w)^k)$  as  $w \rightarrow 0^+$ .

By Lemma 2.2,  $\mathcal{O}_{\beta,k} \subseteq \mathcal{O}_{\beta',k'}$  whenever  $\beta > \beta'$ , or  $\beta = \beta'$  and  $k \leq k'$ ; the class with the smaller exponent, and at equal exponents the larger degree, is the coarser one. A block  $w^\beta P(L)$  with  $\deg P = k$  belongs to  $\mathcal{O}_{\beta,k}$  and to no  $\mathcal{O}_{\beta,k'}$  with  $k' < k$ . Sums and products of remainder classes obey

$$\mathcal{O}_{\beta,k} + \mathcal{O}_{\beta',k'} = \mathcal{O}_{\min(\beta,\beta'),k''}, \quad \mathcal{O}_{\beta,k} \cdot \mathcal{O}_{\beta',k'} = \mathcal{O}_{\beta+\beta',k+k'}, \quad (2.1)$$

where  $k''$  is the degree attached to the smaller exponent, and the larger of the two when the exponents coincide.

## 2.2 Exponent semigroups

**Lemma 2.4** (Local finiteness). Let  $\delta_1, \dots, \delta_m > 0$  and  $S = \{\mathbf{k} \cdot \boldsymbol{\delta} : \mathbf{k} \in \mathbb{N}^m\}$ . For every real  $H$  the set  $\{\mathbf{k} \in \mathbb{N}^m : \mathbf{k} \cdot \boldsymbol{\delta} \leq H\}$  is finite, hence  $S \cap (-\infty, H]$  is finite, and every  $\sigma \in S$  has only finitely many representations  $\sigma = \mathbf{k} \cdot \boldsymbol{\delta}$ .

*Proof.* If  $\mathbf{k} \cdot \boldsymbol{\delta} \leq H$  then  $k_i \leq H/\delta_i$  for each  $i$ . □

When the gaps are incommensurable the *group* generated by them is dense in  $\mathbb{R}$  (for instance  $\mathbb{Z} + \sqrt{2}\mathbb{Z}$ ), but the semigroup of nonnegative combinations has only finitely many elements below any bound. This is the whole reason why expansions with irrational exponents can be computed term by term: the support of every expansion in this article lies in a translate  $\beta_0 + S$  of such a semigroup. Following [11], a subset of  $\mathbb{R}$  that is finite below every bound is called *left-finite*; the semigroups  $S$  are left-finite, and so are their translates and finite unions.

## 2.3 Jets and their arithmetic

**Definition 2.5** (Jets). Fix gaps as above and  $\beta_0 \in \mathbb{R}$ . A *jet* (of cutoff  $H$ ) is a finite sum  $U = \sum_{\sigma \in T} w^\sigma C_\sigma(L)$  with  $T \subset \beta_0 + S$  finite,  $C_\sigma \in \mathbb{R}[L]$ , all  $\sigma < H$ ; equal exponents are merged and zero polynomials are discarded. Its *valuation*  $\text{val } U$  is its least exponent ( $+\infty$  for the empty jet). For nonnegative exponents and  $H > 0$ , the jet algebra  $\mathcal{P}_{<H}$  uses addition and multiplication followed by truncation below  $H$ .

In that nonnegative-exponent ring, blocks of exponent  $\geq H$  form an ideal and  $\mathcal{P}_{<H}$  is the corresponding quotient. This quotient description does not apply to arbitrary real exponents: multiplication by a negative power can move a discarded term below the cutoff. General shifted jets therefore require precision tracking; a factor of valuation  $\gamma < 0$  lowers an unknown exponent cutoff by  $|\gamma|$  (Section 3.2). In the package a jet is a sorted list of pairs  $\{\sigma, C_\sigma\}$  with  $\sigma$  exact and  $C_\sigma$  a polynomial in a private symbol  $\ell$ . Algebraic exponents have canonical `RootReduce` forms; more general exact exponents require provable equality and order comparisons (Section 13).

## Differentiation

**Lemma 2.6** (Euler identity). For  $\beta \in \mathbb{R}$ ,  $P \in \mathbb{R}[L]$  and  $r \in \mathbb{N}$ ,

$$\frac{d}{dw}[w^\beta P(L)] = w^{\beta-1}(\beta + \mathcal{D})P(L), \quad \frac{d^r}{dw^r}[w^\beta P(L)] = w^{\beta-r} \prod_{j=0}^{r-1} (\beta - j + \mathcal{D})P(L). \quad (2.2)$$



The operators  $\beta - j + \mathcal{D}$  commute, and  $(\beta + \mathcal{D})$  preserves  $\deg P$  when  $\beta \neq 0$  and lowers it by one when  $\beta = 0$ .

*Proof.*  $\frac{d}{dw}[w^\beta P(\log w)] = \beta w^{\beta-1}P + w^\beta P'(\log w)/w$ . Iterate.  $\square$

The identity says that differentiation acts on a block as the scalar  $\beta$  plus  $\mathcal{D}$  on the polynomial; the exponent bookkeeping and the polynomial arithmetic separate completely. In the coefficient formulas below operators  $\mathcal{D} + c$  appear with real  $c$  (not necessarily positive), and their action on the coefficient vector  $(z_0, \dots, z_d)$  of  $P = \sum z_s L^s$  is bidiagonal:  $(c + \mathcal{D})P = \sum_s (cz_s + (s + 1)z_{s+1})L^s$ .

## Unit series

Let  $U$  be a jet with  $\text{val } U > 0$ . Then  $U^k$  has valuation  $\geq k \text{val } U$ , so in  $\mathcal{P}_{<H}$  only finitely many powers of  $U$  are nonzero, and every power series  $\Phi(t) = \sum_k \phi_k t^k$  can be substituted:

$$\Phi(U) = \sum_{k < H/\text{val } U} \phi_k U^k \in \mathcal{P}_{<H}. \quad (2.3)$$

In particular  $(1 + U)^a = \sum_k \binom{a}{k} U^k$  for every real  $a$  (with  $\binom{a}{k} = a(a-1)\cdots(a-k+1)/k!$ ),  $\log(1 + U) = \sum_{k \geq 1} (-1)^{k+1} U^k / k$ ,  $e^U = \sum_k U^k / k!$ , and  $\phi(c + U) = \sum_k \phi^{(k)}(c) U^k / k!$  for any  $\phi$  analytic at  $c$ . These are identities between finite sums of blocks modulo exponents  $\geq H$ ; their analytic content is discussed in Section 7.

## Composition with the logarithm

The one operation that mixes the exponent bookkeeping with the polynomial coefficients is the substitution of a jet into the logarithm: if  $w$  is replaced by  $w(1 + U)$  then  $L$  becomes  $L + \log(1 + U)$ , and a block  $w^a P(L)$  becomes  $w^a (1 + U)^a P(L + \log(1 + U))$ . The following recurrence computes this without expanding  $\log(1 + U)$  separately.

**Lemma 2.7** (Composition recurrence). *Let  $P \in \mathbb{R}[L]$  and  $a \in \mathbb{R}$ . For  $|t| < 1$ ,*

$$(1 + t)^a P(L + \log(1 + t)) = \sum_{k \geq 0} Q_k(L) t^k, \quad Q_0 = P, \quad Q_{k+1} = \frac{(a - k)Q_k + Q'_k}{k + 1}, \quad (2.4)$$

with  $Q_k \in \mathbb{R}[L]$ ,  $\deg Q_k \leq \deg P$ , and, when  $a \in \mathbb{N}$  and  $P$  is constant,  $Q_k = 0$  for  $k > a$ . Consequently, for a jet  $U$  of positive valuation,  $(1 + U)^a P(L + \log(1 + U)) = \sum_k Q_k(L) U^k$  in  $\mathcal{P}_{<H}$ .

*Proof.* Put  $F(t) = (1 + t)^a P(L + \log(1 + t))$ ; it is analytic in  $|t| < 1$  with values in  $\mathbb{R}[L]$ , so  $F = \sum Q_k t^k$  with  $Q_k \in \mathbb{R}[L]$  and  $Q_0 = P$ . Differentiating,  $(1 + t)\partial_t F = aF + \partial_L F$ , because  $\partial_t$  acting on the two factors gives  $a(1 + t)^{a-1}P + (1 + t)^{a-1}P'$ . Comparing coefficients of  $t^k$  in  $(1 + t)\sum (k + 1)Q_{k+1}t^k = \sum (aQ_k + Q'_k)t^k$  gives  $(k + 1)Q_{k+1} + kQ_k = aQ_k + Q'_k$ . The degree does not increase, and for  $a \in \mathbb{N}$  and constant  $P$  the factor  $(a - k)$  annihilates  $Q_{a+1}$  and hence all later  $Q_k$  (for nonconstant  $P$  the derivative keeps the sequence alive:  $\log(1 + t)$  is not a polynomial). The jet statement follows because substitution of  $U$  for  $t$  is a ring homomorphism on finite sums.  $\square$

*Remark 2.8.* The recurrence is the whole composition calculus needed here: the forward model  $ax^p(1 + \sum x^{\delta_i} B_i(\log x))$  evaluated at  $x = z(1 + U)$  is  $az^p(1 + U)^p(1 + \sum_i z^{\delta_i}(1 + U)^{\delta_i} B_i(L + \log(1 + U)))$ , each factor of which is an instance of (2.4) (with  $P = 1$  for the first). Its derivative is obtained from the same lemma with  $a$  replaced by  $a - 1$  and  $P$  by  $aP + P'$ , since  $\frac{d}{dx}[x^a P(\log x)] = x^{a-1}(aP + P')(\log x)$ .



## 2.4 Blocks with nonconstant leading coefficient

Everything above allows the polynomial in the leading block to be nonconstant. In general, a real non-integer power or a logarithm of such a block produces  $P(L)^a$  or  $\log P(L)$  outside  $\mathbb{R}[L]$ . Special algebraic simplifications may stay in the class, but generic expressions require larger scales (rational functions of  $L$ , or iterated logarithms, Section 12). Nonnegative integer powers of any jet stay in the class.

## 3 Forward expansions

Before inverting anything we need the forward function in the normal form  $ax^p(1 + \sum_i x^{\delta_i} B_i(\log x))$ , and the second question [2] is a forward problem in its own right. This section shows that a large class of expressions has convergent power-log expansions, and describes the precision-tracking evaluator that computes them with a rigorous remainder class.

### 3.1 Convergent power-log expansions

**Definition 3.1.** A function  $F$  on  $(0, w_0)$  has a *convergent power-log expansion* with support in  $\beta_0 + S$  ( $S$  a finitely generated positive semigroup with gaps  $\delta_1, \dots, \delta_m$ ) if there are polynomials  $C_\sigma \in \mathbb{R}[L]$ , constants  $M, K \geq 1$ , and integers  $d_0, d \geq 0$  such that

$$F(w) = \sum_{\sigma \in S} w^{\beta_0 + \sigma} C_\sigma(L) \quad \text{for } 0 < w < w_1, \quad \|C_\sigma\|_1 \leq MK^{n(\sigma)}, \quad \deg C_\sigma \leq d_0 + dn(\sigma), \quad (3.1)$$

where  $n(\sigma)$  is the least  $|\mathbf{k}|$  with  $\mathbf{k} \cdot \boldsymbol{\delta} = \sigma$ ,  $\|P\|_1$  is the sum of the absolute values of the coefficients of  $P$ , and the series converges absolutely. Thus  $|C_\sigma(L)| \leq MK^{n(\sigma)} M(w)^{d_0 + dn(\sigma)}$ . Zero coefficients satisfy the degree bound by convention. We write  $F \in \mathcal{E}$ . The fixed allowance  $d_0$  is essential: it admits a polynomial in  $\log w$  at the leading exponent, including  $\log w$  itself.

The coefficient bound makes the tail after any cutoff controllable (Lemma 3.2 below), and it is stable under all the operations we need.

**Lemma 3.2** (Tail of a convergent expansion). *Let  $F \in \mathcal{E}$  as in (3.1),  $\delta_* = \min_i \delta_i$ , and  $H > 0$ . Then*

$$F(w) - \sum_{\sigma < H} w^{\beta_0 + \sigma} C_\sigma(L) = \mathcal{O}(w^{\beta_0 + H} M(w)^{d_0 + Nd}), \quad N = \lceil H/\delta_* \rceil. \quad (3.2)$$

*More precisely, the omitted part equals  $w^{\beta_0 + \sigma_*} C_{\sigma_*}(L) + o(w^{\beta_0 + \sigma_*})$  where  $\sigma_* = \min\{\sigma \in S : \sigma \geq H\}$ .*

*Proof.* Split the omitted terms into those with  $n(\sigma) \leq N - 1$  and the rest. The first group is finite (Lemma 2.4) and each term is  $\mathcal{O}(w^{\beta_0 + H} M^{d_0 + (N-1)d})$ . For the rest, the number of weights whose shortest representation has length  $n$  is at most  $m^n$ , and  $\sigma \geq n\delta_*$ , so their sum is bounded by

$$Mw^{\beta_0} M^{d_0} \sum_{n \geq N} (mKw^{\delta_*} M^d)^n = \mathcal{O}(w^{\beta_0 + N\delta_*} M^{d_0 + Nd}).$$

Since  $N\delta_* \geq H$ , this proves the bound. For the refined statement let  $\sigma_{**}$  be the next element of  $S$  after  $\sigma_*$ , which exists by local finiteness when  $S$  has a generator, and choose  $\sigma_* < H' < \sigma_{**}$ . Applying the bound at  $H'$  shows that everything beyond  $\sigma_*$  is  $\mathcal{O}(w^{\beta_0 + H'} M^{d_0 + N'd}) = o(w^{\beta_0 + \sigma_*})$ . The difference  $\sigma_{**} - \sigma_*$  need not be at least  $\delta_*$ . Finite support with no further terms has zero tail.  $\square$

**Theorem 3.3** (Closure). *The class  $\mathcal{E}$  contains the constants and  $w$  and is closed under sums, products, and the following operations, with the supports and gaps indicated.*

1. *If  $F \in \mathcal{E}$  has leading block  $cw^\beta$  with a constant  $c > 0$ , then  $F^a \in \mathcal{E}$  for every real  $a$  and  $\log F \in \mathcal{E}$ ; their supports are contained in  $a\beta + S'$  and  $\{0\} \cup S'$  respectively, with block  $\log c + \beta L$  at exponent 0 in the logarithm. Here  $S'$  is a finitely generated positive semigroup containing the exponents of  $F/(cw^\beta) - 1$ ; the proof establishes that such a semigroup exists.*
2. *If  $F \in \mathcal{E}$  has no negative-exponent blocks and its exponent-0 block is  $c + kL$  with  $k \in \mathbb{R}$ , then  $e^F \in \mathcal{E}$  with leading block  $e^c w^k$ . The argument  $F$  may grow logarithmically.*
3. *If  $F \in \mathcal{E}$  has  $\beta_0 \geq 0$  and constant exponent-0 block  $c$ , and  $\phi$  is analytic at  $c$ , then  $\phi \circ F \in \mathcal{E}$ .*

*Proof.* First note that a positive-valuation expansion  $U$  can be represented with support in a positive semigroup based at 0: if necessary, add its positive leading exponent as a generator and enlarge the constants in (3.1). Choose one shortest multi-index  $\mathbf{k}$  for each resulting weight  $\sigma$ , put  $D = d_0 + d$ , and introduce

$$T_{ij} = w^{\delta_i} L^j, \quad 1 \leq i \leq m, \quad 0 \leq j \leq D.$$

For  $n = |\mathbf{k}| \geq 1$ , every  $w^\sigma L^s$  with  $s \leq d_0 + dn \leq Dn$  is a product of  $n$  such parameters: distribute the  $s$  logarithmic factors among the  $n$  generator factors, each receiving at most  $D$ . Fix one such distribution for every monomial. The coefficient-norm bound makes the resulting series analytic in a sufficiently small polydisc, since its terms of degree  $n$  have total absolute coefficient at most  $M(mK)^n$ . Analytic composition with  $(1 + U)^a$ ,  $\log(1 + U)$ ,  $e^U$ , or  $\phi(c + U)$  is therefore legitimate. Cauchy estimates give exponential coefficient bounds in the new parameters, and their logarithmic degree grows at most linearly in parameter degree.

Regrouping by weight preserves bounds of the form (3.1): a representation of weight  $\sigma$  has length at most  $\sigma/\delta_*$ , whereas its shortest length is at least  $\sigma/\delta_{\max}$ ; their ratio is bounded, and the number of parameter monomials of a given total degree is at most exponential. Enlarge  $M, K, d$  to absorb both factors.

For sums and products, use the union of the generator sets and, for a sum of differently shifted supports, the positive difference between their shifts as an additional generator. Coefficient norms are subadditive and submultiplicative, so the same bounds apply. If leading blocks cancel, rebasing at the next nonzero exponent still admits finitely many positive generators: the multi-indices above that exponent form an upward-closed subset of  $\mathbb{N}^m$ , with finitely many coordinatewise minimal elements. Subtracting the new leading exponent from the weights of those elements supplies finitely many nonnegative shifts, together with the original generators. This also justifies normalizing  $F$  by its actual leading monomial in (1). Separate the finite leading polynomial before applying the small-parameter construction; in (2) separate  $c + kL$  to obtain the factor  $e^c w^k$ .  $\square$

The theorem supplies sufficient closure conditions, and every truncation is a genuine asymptotic expansion by Lemma 3.2. They are not necessary conditions for every individual expression: for example  $\sqrt{L^2} = -L$  for  $L < 0$ , and polynomials may be applied to an unbounded logarithm. In general, non-integer powers of a nonconstant leading polynomial, its logarithm, exponentials of negative-power arguments, and oscillatory functions of an unbounded logarithm require larger scales (Section 12).

**Corollary 3.4** (Laurent and Puiseux composition). *If  $U \in \mathcal{E}$  has a positive constant leading coefficient and positive valuation, and a selected real branch of  $\phi(c + v)$  has a convergent Puiseux expansion  $v^{n_0/d} A(v^{1/d})$  for  $v \rightarrow 0^+$ , with  $d \geq 1$  an integer and  $A$  analytic at 0, then  $\phi(c + U) \in \mathcal{E}$ . The same holds from below using  $\phi(c - v)$ .*

*Proof.* Apply the closure theorem to  $U^{1/d}$ , the analytic function  $A$ , and  $U^{n_0/d}$ . Negative  $n_0$  includes poles. The side of approach fixes the real branch of a ramified head.  $\square$

Variable powers also reduce to the closure theorem through  $F^G = \exp(G \log F)$  when  $F > 0$  and the resulting exponential argument satisfies its hypotheses; this includes  $w^w = \exp(w \log w)$ . For a nonzero leading block  $w^\beta P(L)$ , the eventual sign is the sign of  $(-1)^{\deg P} [L^{\deg P}]P$ , so absolute values reduce to multiplication by that sign whenever it is provable.

### 3.2 The precision-tracking evaluator

The package computes forward expansions by structural recursion over the expression tree, carrying for every subexpression a triple  $(T, P, D)$ : a jet  $T$  and the assertion

$$\text{subexpression} = T + \mathcal{O}(w^P M(w)^D), \quad P \in \mathbb{R} \cup \{+\infty\}, \quad (3.3)$$

where  $P = +\infty$  means that  $T$  is exact. A global working order  $K_w$  bounds the number of terms of every unit series that is expanded. The rules, all justified by (2.1) and Lemma 3.2, are:

- **constants** and  $w$  are exact; **Plus**: compute  $T_1 + T_2$  and truncate below  $P = \min(P_1, P_2)$ . Combine the input remainder degrees at  $P$  with the degree of every explicit block discarded at exponent  $P$ ;
- **Times**: if  $R_i$  denotes the input remainder, the omitted terms include  $R_1 T_2$ ,  $R_2 T_1$  and  $R_1 R_2$ . Their exponent/degree pairs follow from (2.1); use the coarsest pair and truncate  $T_1 T_2$  below that exponent. Also include the degrees of explicit product blocks discarded exactly at this boundary. A nonempty jet has valuation strictly below its remainder exponent; an empty jet contributes its remainder's exponent and degree when estimating its size;
- **Power** with an exact real exponent  $a$ : nonnegative integers by binary exponentiation with precision tracked at each product (exact for exact input). Otherwise the leading block  $cw^\beta$  of  $T$  must have constant  $c > 0$  (or  $c < 0$  with integer  $a$ ); with  $U = T/(cw^\beta) - 1$ , relative precision  $P - \beta$  and relative cutoff  $c_{\text{rel}} = \min(P - \beta, K_w - a\beta)$ , the result is  $c^a w^{a\beta} \sum_k \binom{a}{k} U^k$  with precision  $a\beta + c_{\text{rel}}$ . For a nonempty  $U$ , its series-truncation degree is bounded by  $\lceil c_{\text{rel}} / \text{val } U \rceil \deg_{\text{max}} U$ , combined with the input error degree  $D$ ; an empty  $U$  has no computed series tail;
- **Log**: the leading block must be  $cw^\beta$  with constant  $c > 0$ ; the result is  $\log c + \beta L + \log(1 + U)$ ;
- **Exp**:  $T$  must have no negative exponents and its exponent-0 block must be  $c + kL$  with exact numeric  $k$ ; the result is  $e^{cw^k} \exp(U)$ ;
- an analytic head  $\phi$  (**Sin**, **ArcTan**, **BesselJ**[0, ·], ...) applied to  $T = c + U$  with constant  $c$ :  $\sum_k \phi^{(k)}(c) U^k / k!$  with the Taylor coefficients obtained from **Series**[ $\phi[c + \mathbf{t}]$ , {**t**, 0, N}];
- any other subexpression is expanded by **Series**[**expr**, {**w**, 0, **n**}, **Assumptions**  $\rightarrow \mathbf{w} > 0$ ]; the coefficients (which may contain  $\log w$ ) and the irrational-power monomials that **Series** leaves outside its **SeriesData** are parsed as jets, the precision is the order of the 0 term, and the logarithmic degree  $D$  is read from the first omitted block obtained by expanding one order further.

The whole expression is re-expanded with a larger  $K_w$  until the final precision reaches the requested cutoff. The result is reported as the retained blocks plus  $\mathcal{O}_{\sigma_*, k_*}$ , where  $\sigma_*$  is the exponent of the first omitted block if one lies below the final precision (its complete coefficient having been computed), and otherwise the precision pair  $(P, D)$  itself.

**Proposition 3.5** (Soundness). *If every leaf and every Taylor, Laurent, Puiseux or fallback expansion supplied by **Series** satisfies its asserted branch and remainder contract, then the root satisfies (3.3), and the reported remainder class is valid.*

*Proof.* Induct over the expression tree. For sums and products, use (2.1) for propagated errors and add the explicit discarded blocks to the remainder; in particular, their logarithmic degrees at the precision boundary cannot be ignored. For unit series, apply Lemma 3.2 to the finite jet's positive generators and the mean value estimate  $\Phi(1 + U) - \Phi(1 + U_{\text{known}}) = \mathcal{O}(U - U_{\text{known}})$  for its unknown part. Laurent and Puiseux heads are reduced to these same operations on a known signed increment. The final remainder statement follows from the refined tail bound whenever a complete first omitted block has been computed below the precision limit.  $\square$

When **Series** supplies a Taylor, Laurent, Puiseux or fallback expansion, its branch and remainder assertions remain inputs to the soundness proposition. Observing the degree of one omitted block alone does not prove that an arbitrary unknown tail satisfies the resulting bound. The structural operations preserve valid assertions, but the package does not prove every analytic contract of a special function or a user-supplied remainder. If cancellation has hidden a leading increment needed by a pole or branch point, the working precision must increase before composition.

### 3.3 The second question

For  $F(x) = (1 + x + x^{\sqrt{2}})^{\sqrt{2}}$  at  $x \rightarrow +\infty$  the local variable is  $w = 1/x$  and

$$F = w^{-2}(1 + w^{\sqrt{2}-1} + w^{\sqrt{2}})^{\sqrt{2}} = w^{-2} \sum_{k \geq 0} \binom{\sqrt{2}}{k} (w^{\sqrt{2}-1} + w^{\sqrt{2}})^k,$$

an instance of item (1) of Theorem 3.3 with gaps  $\sqrt{2} - 1$  and  $\sqrt{2}$ . The exponents of  $x$  are  $2 - (\sqrt{2} - 1)k_1 - \sqrt{2}k_2$ ; ordering the semigroup elements gives (1.5), and the first omitted exponent is  $7 - 5\sqrt{2}$  (multi-index  $(5, 0)$ ). The workaround in the question, which peels off one leading term per limit computation, produces the same seven terms; the evaluator supplies them with the remainder  $\mathcal{O}(x^{7-5\sqrt{2}})$  and can continue to larger finite cutoffs within its resource budget.

## 4 The model and the real branch

### 4.1 The normal form

Let  $u \rightarrow 0^+$  be the local variable (later  $u = x - x_0$ ,  $x_0 - x$ ,  $1/x$  or  $-1/x$ ). The central object is a finite *power-log germ*

$$f(u) = y_0 + a u^p \left( 1 + \sum_{i=1}^m u^{\delta_i} B_i(\log u) \right), \quad a \neq 0, \ p \neq 0, \ \delta_i > 0, \ B_i \in \mathbb{R}[L] \setminus \{0\}, \quad (4.1)$$

with  $y_0 \in \mathbb{R}$  when  $p > 0$  and  $y_0 = 0$  when  $p < 0$  (a constant term is then a correction with gap  $-p$ ). Equal gaps are merged. The *leading block*  $au^p$  has a constant coefficient; a logarithmic factor in the leading block ( $u \log u$ ,  $u(\log u)^{-1}$ ) changes the problem (Section 12). Everything below is stated for this exact finite germ; functions that are only known through a truncated expansion are handled in Section 9.

The *uniformizer* is

$$z = \left(\frac{v}{a}\right)^{1/p}, \quad v = y - y_0 \ (p > 0), \quad v = y \ (p < 0), \quad L = \log z = \frac{\log(v/a)}{p}, \quad (4.2)$$

so that  $z \rightarrow 0^+$  in every case: for  $p > 0$  because  $v \rightarrow 0$  with  $v/a > 0$ , for  $p < 0$  because  $v \rightarrow \pm\infty$  with the sign of  $a$ . The role of  $z$  is that the inverse satisfies  $u \sim z$ .

## 4.2 Existence and uniqueness of the local inverse

**Theorem 4.1** (Positive local branch). *For the germ (4.1) there is  $\varepsilon > 0$  such that  $f$  is strictly monotone on  $(0, \varepsilon)$ , with*

$$f(u) - y_0 \sim a u^p, \quad f'(u) \sim a p u^{p-1} \quad (u \rightarrow 0^+). \quad (4.3)$$

*Its inverse  $g$ , defined on the image of  $(0, \varepsilon)$ , is the unique function with  $g > 0$ ,  $g \rightarrow 0$  and  $f(g(y)) = y$  there, and it satisfies*

$$g(y) \sim z, \quad g'(y) \sim \frac{z}{p v} \quad (z \rightarrow 0^+). \quad (4.4)$$

*Proof.* Write  $R(u) = \sum_i u^{\delta_i} B_i(\log u)$ . By Lemma 2.2,  $R(u) \rightarrow 0$  and, since  $uR'(u) = \sum_i u^{\delta_i} (\delta_i B_i + B'_i)(\log u)$  by Lemma 2.6, also  $uR'(u) \rightarrow 0$ . Hence

$$f'(u) = a p u^{p-1} \left(1 + R(u) + \frac{uR'(u)}{p}\right) = a p u^{p-1} (1 + o(1)),$$

which has the sign of  $ap$  on some  $(0, \varepsilon)$ ; this gives monotonicity and (4.3). The inverse exists on the image, tends to 0 because  $f$  is a homeomorphism of  $(0, \varepsilon)$  onto its image, and is the only solution of  $f(u) = y$  in  $(0, \varepsilon)$ . Substituting  $u = g(y)$  into  $f(u) - y_0 \sim a u^p$  gives  $v \sim a g(y)^p$ , i.e.  $g \sim z$ ; and  $g' = 1/f'(g) \sim 1/(ap g^{p-1}) = g/(p a g^p) \sim z/(p v)$ .  $\square$

The theorem is local. Whether the local branch extends to a global inverse is a separate question, answered for the two examples of [1] as follows.

**Proposition 4.2** (Global invertibility of the examples).  *$f_1(x) = x + x^2(1 + \log x)$  and  $f_2(x) = x + x^{\sqrt{2}}$  are strictly increasing bijections of  $(0, \infty)$  onto itself, with*

$$f'_1(x) = 1 + x(3 + 2 \log x) \geq m_1 := 1 - 2e^{-5/2} > 0.835, \quad f'_2(x) = 1 + \sqrt{2} x^{\sqrt{2}-1} > 1. \quad (4.5)$$

*Their inverses  $g_1, g_2$  are  $C^1$  up to  $y = 0$  with  $g_i(0) = 0$ ,  $g'_i(0) = 1$ , but  $g_1$  is not  $C^2$  at 0:  $g''_1(y) = -f''_1(g_1)/f'_1(g_1)^3 \sim -(5 + 2 \log y) \rightarrow +\infty$ .*

*Proof.*  $x(3 + 2 \log x)$  has derivative  $5 + 2 \log x$ , minimum at  $x = e^{-5/2}$  with value  $-2e^{-5/2}$ ; the limits of  $f_1$  at 0 and  $\infty$  are 0 and  $\infty$ . The rest is immediate.  $\square$

So for both examples the branch of Theorem 4.1 is *the* real inverse, and the expansions (1.3)–(1.4) describe it near 0; the absence of a second-order Taylor expansion of  $g_1$  at 0 is exactly why  $\log y$  must appear at  $y^2$ .

### 4.3 The complex germ returned by `InverseFunction`

The equation  $f_1(x) = 0$ ,  $x \neq 0$ , reads  $1 + x(1 + \log x) = 0$ . With  $t = 1 + \log x$  it becomes  $te^t = -e$ , so  $t = W_k(-e)$  and

$$c_k = e^{W_k(-e)-1} = -\frac{1}{W_k(-e)}, \quad f'_1(c_k) = c_k - 1, \quad f''_1(c_k) = 3 + 2W_k(-e), \quad (4.6)$$

where the derivative identities follow from  $c_k W_k(-e) = -1$ , provided the logarithm near  $c_k$  is chosen to have value  $W_k(-e) - 1$  there. This branch qualification matters: with the fixed principal logarithm used by `Log`, the relevant roots are the conjugate pair  $k = 0, -1$ ; other  $k$  require other local logarithm branches. For each compatible choice the derivative is nonzero, and the analytic inverse function theorem gives an inverse germ through  $c_k$ . The reported `Series[InverseFunction[...][y], {y, 0, 2}]` result chooses  $k = -1$ :

$$c + \frac{y}{c-1} - \frac{f''_1(c)}{2f'_1(c)^3}y^2 + \cdots = c + \frac{y}{c-1} - \frac{3+2W_{-1}(-e)}{2(c-1)^3}y^2 + \mathcal{O}(y^3),$$

which is (1.2). Nothing is wrong with this answer: it is the inverse of a different branch of  $f_1^{-1}$ , the one through a complex root, and adding `Assumptions -> y > 0` cannot repair it, because the reality of  $y$  says nothing about which preimage is intended. The real branch is characterised by  $g(y) \rightarrow 0$ , and that condition, not an assumption on  $y$ , is what an inversion routine must be given; the package builds the branch of Theorem 4.1 directly and never asks `InverseFunction` to choose.

## 5 Lagrange–Bürmann inversion with a marker

The classical Lagrange inversion theorem expands the inverse of an analytic map at a point where its derivative does not vanish. Our maps are not analytic at 0, and the inverse has no Taylor series there; but at every *positive* point the map is analytic, and the trick is to invert a perturbation  $u + \eta h(u) = q$  around the positive point  $q$  in the perturbation parameter  $\eta$ . The logarithms and irrational powers in  $h$  are then analytic functions of  $u$  near  $q > 0$ , and the derivatives that appear are computed by the Euler identity without ever evaluating anything at 0.

### 5.1 The marker theorem

**Theorem 5.1** (Lagrange–Bürmann with a marker). *Let  $q > 0$  and let  $h$  and  $\Phi$  be analytic in a complex neighbourhood of  $q$ . For  $|\eta|$  sufficiently small the equation*

$$u + \eta h(u) = q \quad (5.1)$$

*has a unique solution  $u(\eta)$  near  $q$  with  $u(0) = q$ , it is analytic in  $\eta$ , and*

$$\Phi(u(\eta)) = \Phi(q) + \sum_{N \geq 1} \frac{(-\eta)^N}{N!} \frac{d^{N-1}}{dq^{N-1}} [\Phi'(q) h(q)^N]. \quad (5.2)$$

*The same identity holds coefficientwise in  $\eta$  for arbitrary formal power series  $h, \Phi$  with coefficients in any commutative  $\mathbb{Q}$ -algebra, i.e. as an identity of formal power series in  $\eta$  whose coefficients are polynomials in finitely many derivatives of  $h$  and  $\Phi$  at  $q$ .*

*Proof.* Choose  $\rho > 0$  so small that  $h$  and  $\Phi$  are analytic on the closed disc  $|s - q| \leq \rho$ , and let  $M = \max_{|s-q|=\rho} |h(s)|$ . For  $|\eta| < \rho/M$  we have  $|\eta h(s)| < |s - q|$  on the circle  $C = \{|s - q| = \rho\}$ ,

so by Rouché's theorem  $F_\eta(s) = s - q + \eta h(s)$  has exactly one zero  $u(\eta)$  in the disc, and by the argument principle

$$\Phi(u(\eta)) = \frac{1}{2\pi i} \oint_C \Phi(s) \frac{F'_\eta(s)}{F_\eta(s)} ds = \frac{1}{2\pi i} \oint_C \Phi(s) \frac{1 + \eta h'(s)}{s - q + \eta h(s)} ds.$$

On  $C$  the geometric series  $\frac{1}{s - q + \eta h(s)} = \sum_{N \geq 0} \frac{(-\eta)^N h(s)^N}{(s - q)^{N+1}}$  converges uniformly, so

$$\Phi(u(\eta)) = \sum_{N \geq 0} (-\eta)^N \left[ \frac{1}{2\pi i} \oint_C \frac{\Phi(s) h(s)^N}{(s - q)^{N+1}} ds + \frac{1}{2\pi i} \oint_C \frac{\eta \Phi(s) h'(s) h(s)^N}{(s - q)^{N+1}} ds \right].$$

By Cauchy's formula the first integral is  $\frac{1}{N!} \frac{d^N}{dq^N} [\Phi h^N](q)$  and the second, after shifting the index  $N \rightarrow N - 1$ , contributes  $-\frac{1}{(N-1)!} \frac{d^{N-1}}{dq^{N-1}} [\Phi h' h^{N-1}](q)$  to the coefficient of  $(-\eta)^N$  for  $N \geq 1$ . Since  $\frac{d}{dq} [\Phi h^N] = \Phi' h^N + N \Phi h' h^{N-1}$ , the coefficient of  $(-\eta)^N$  is

$$\frac{1}{N!} \frac{d^{N-1}}{dq^{N-1}} [\Phi' h^N + N \Phi h' h^{N-1}] - \frac{1}{(N-1)!} \frac{d^{N-1}}{dq^{N-1}} [\Phi h' h^{N-1}] = \frac{1}{N!} \frac{d^{N-1}}{dq^{N-1}} [\Phi' h^N],$$

which is (5.2); the constant term is  $\Phi(q)$ . Analyticity of  $u(\eta)$  follows from the implicit function theorem at  $(q, 0)$ , where  $\partial_u F_0 = 1$ . For the formal statement, both sides of (5.2) have coefficients that are universal polynomials in the Taylor coefficients of  $h$  and  $\Phi$  at  $q$ ; two polynomials that agree for all analytic data agree identically (they agree on a Zariski-dense set), so the identity holds for arbitrary formal coefficients.  $\square$

*Remark 5.2.* Other proofs exist [4, 5, 3]; the one above is chosen because it exhibits the analytic content: the solution is the unique root in a disc around  $q$ , a fact reused in Section 7 to obtain explicit constants. The formal statement is what makes the coefficient formula of Section 6 an identity, independent of any convergence question.

## 5.2 Consequences

**Near-identity form.** With  $\Phi(q) = q$  and  $\eta = 1$ ,

$$f(u) = u + h(u) \implies g(y) = y + \sum_{N \geq 1} \frac{(-1)^N}{N!} \frac{d^{N-1}}{dy^{N-1}} [h(y)^N], \quad (5.3)$$

formally; Sections 7 and 9 say when setting  $\eta = 1$  is legitimate as an asymptotic statement. The formula (5.3) is a complete answer for the log example: with  $h(y) = y^2(1 + \log y)$ , the  $N$ -th term is  $\frac{(-1)^N}{N!} \partial_y^{N-1} [y^{2N}(1 + \log y)^N]$ , computed by Lemma 2.6; for  $N = 2$ ,  $\frac{1}{2} \partial_y [y^4(1 + L)^2] = y^3(2(1 + L)^2 + (1 + L)) = y^3(2L^2 + 5L + 3)$ .

**Several markers.** If  $h = \sum_i \eta_i h_i$  with independent markers, expanding  $h^N$  multinomially in (5.2) gives the coefficient of  $\prod \eta_i^{k_i}$ ,  $|\mathbf{k}| = N$ , as

$$\frac{(-1)^N}{\mathbf{k}!} \frac{d^{N-1}}{dq^{N-1}} \left[ \Phi'(q) \prod_i h_i(q)^{k_i} \right], \quad (5.4)$$

because the multinomial coefficient  $N!/\mathbf{k}!$  cancels the  $1/N!$ .

**Transport of observables.** Formula (5.2) gives at once the expansion of any analytic function of the inverse. With  $\Phi(u) = u^r$  one obtains powers of the inverse; with  $\Phi = \log$  its logarithm; differentiating the identity for  $u^r$  with respect to  $r$  at  $r = 0$  gives the logarithm again. These are used in Section 6.



**General core.** Suppose  $F = F_0 + R$  where  $F_0$  has a known inverse  $\phi$  on the branch of interest and  $R$  is a perturbation. Put  $t = F_0(x)$ , so that  $F(x) = y$  becomes  $t + \eta R(\phi(t)) = y$  with  $\eta = 1$ ; Theorem 5.1 with  $\Phi = \phi$  yields

$$x = \phi(y) + \sum_{N \geq 1} \frac{(-1)^N}{N!} \frac{d^{N-1}}{dy^{N-1}} [\phi'(y) R(\phi(y))^N]. \quad (5.5)$$

This is the “phase-coordinate” inversion of [11]; it is implemented as `PerturbativeInverse`[ $\phi$ ,  $R$ ,  $\{x, y\}$ ,  $n$ ] and is the natural tool when the dominant part of  $F$  is not a monomial (Section 12). For the monomial core  $F_0(x) = ax^p$ ,  $\phi(y) = (y/a)^{1/p}$ , it is equivalent to the substitution  $t = x^p$  used in the next section.

## 6 The coefficient theorem

### 6.1 Statement

Let  $f$  be the germ (4.1),  $z$  the uniformizer (4.2),  $L = \log z$ ,  $\mathcal{D} = d/dL$ . For a multi-index  $\mathbf{k} \in \mathbb{N}^m$  write  $n = |\mathbf{k}|$ ,  $\sigma = \mathbf{k} \cdot \boldsymbol{\delta}$  and  $B^{\mathbf{k}} = \prod_i B_i^{k_i}$ .

**Theorem 6.1** (Euler-operator coefficient formula). *For every real  $r$ , the branch  $g$  of Theorem 4.1 satisfies, formally and (Theorem 7.1) as a convergent expansion for small  $z$ ,*

$$g(y)^r = z^r + \sum_{\mathbf{k} \neq 0} z^{r+\mathbf{k} \cdot \boldsymbol{\delta}} C_{\mathbf{k}}^{(r)}(L), \quad C_{\mathbf{k}}^{(r)}(L) = \frac{(-1)^n r}{p^n \mathbf{k}!} \prod_{j=1}^{n-1} (\mathcal{D} + r + \sigma + pj) B^{\mathbf{k}}(L). \quad (6.1)$$

The empty product ( $n = 1$ ) is the identity. Multi-indices of equal weight are summed: the block of  $g^r$  at  $z^{r+\sigma}$  is  $C_{\sigma}^{(r)} = \sum_{\mathbf{k} \cdot \boldsymbol{\delta} = \sigma} C_{\mathbf{k}}^{(r)}$ . In particular

$$g(y) = z \left( 1 + \sum_{\mathbf{k} \neq 0} z^{\mathbf{k} \cdot \boldsymbol{\delta}} C_{\mathbf{k}}(L) \right), \quad C_{\mathbf{k}} = C_{\mathbf{k}}^{(1)}, \quad (6.2)$$

$$\log g(y) = L + \sum_{\mathbf{k} \neq 0} z^{\mathbf{k} \cdot \boldsymbol{\delta}} \frac{(-1)^n}{p^n \mathbf{k}!} \prod_{j=1}^{n-1} (\mathcal{D} + \sigma + pj) B^{\mathbf{k}}(L).$$

*Proof.* Put  $t = u^p$  and  $q = v/a$  (so  $q = z^p$ ). The equation  $f(u) = y$  becomes

$$q = t + \sum_i t^{1+\delta_i/p} B_i \left( \frac{\log t}{p} \right),$$

which is (5.1) with markers:  $t + \sum_i \eta_i h_i(t) = q$ ,  $h_i(t) = t^{1+\delta_i/p} B_i(\log t/p)$ , at  $\eta_i = 1$ . The observable is  $\Phi(t) = t^{r/p} = u^r$ . All  $h_i$  and  $\Phi$  are analytic near any  $q > 0$ . By (5.4) the coefficient of  $\prod \eta_i^{k_i}$  is

$$\frac{(-1)^n}{\mathbf{k}!} \frac{d^{n-1}}{dq^{n-1}} \left[ \frac{r}{p} q^{r/p-1} \prod_i (q^{1+\delta_i/p} B_i(\frac{\log q}{p}))^{k_i} \right] = \frac{(-1)^n r}{p \mathbf{k}!} \frac{d^{n-1}}{dq^{n-1}} \left[ q^{n-1+(r+\sigma)/p} B^{\mathbf{k}}(\frac{\log q}{p}) \right].$$

Apply the repeated Euler identity (2.2) in the variable  $q$ , with  $\Lambda = \log q$  and the polynomial  $\tilde{P}(\Lambda) = B^{\mathbf{k}}(\Lambda/p)$ : the derivative of order  $n-1$  of  $q^b \tilde{P}(\Lambda)$ ,  $b = n-1+(r+\sigma)/p$ , is  $q^{b-n+1} \prod_{j=0}^{n-2} (b-j + d/d\Lambda) \tilde{P}$ . Since  $L = \Lambda/p$ ,  $d/d\Lambda = \mathcal{D}/p$ , and  $b-j = (r+\sigma + p(n-1-j))/p$ ; hence each factor is  $\frac{1}{p} (\mathcal{D} + r + \sigma + p(n-1-j))$ , and with  $j' = n-1-j$  running from 1 to  $n-1$  the product is  $p^{-(n-1)} \prod_{j'=1}^{n-1} (\mathcal{D} + r + \sigma + pj') B^{\mathbf{k}}(L)$ . Finally  $q^{b-n+1} = q^{(r+\sigma)/p} = z^{r+\sigma}$ , and the

prefactor  $r/p$  combines with  $p^{-(n-1)}$  into  $r/p^n$ . The formal validity is Theorem 5.1's formal clause, applied to the finitely many markers; setting all  $\eta_i = 1$  is legitimate coefficientwise because by Lemma 2.4 only finitely many multi-indices contribute to any exponent. The formula for  $\log g$  is the derivative of (6.1) with respect to  $r$  at  $r = 0$ , where the factor  $r$  vanishes and only the term with the factor differentiated survives; equivalently it is Theorem 5.1 with  $\Phi = \log$ .  $\square$

*Remark 6.2.* The formula (6.1) is the one on which all nine reports of Section 14 agree, with the same normalization: sign  $(-1)^n$ , the multi-index factorial  $\mathbf{k}!$  (not  $n!$ : the multinomial coefficient from expanding  $h^n$  has been absorbed), the factor  $p^{-n}$  (one  $1/p$  from  $\Phi'$  and one from each of the  $n - 1$  derivatives, through  $d/d\Lambda = \mathcal{D}/p$ ), and the operator constants  $r + \sigma + pj$ . Forgetting the substitution  $L = \log(v/a)/p$  in the coefficients, or the factor  $p^{-n}$ , are the two classical mistakes for nonunit  $p$ .

## 6.2 First orders and cross terms

For a single block ( $m = 1$ , gap  $\delta$ , polynomial  $B$ ):

$$g = z - \frac{z^{1+\delta}}{p}B + \frac{z^{1+2\delta}}{2p^2}((1+2\delta+p)B^2 + 2BB') - \frac{z^{1+3\delta}}{6p^3}(\mathcal{D}+1+3\delta+p)(\mathcal{D}+1+3\delta+2p)B^3 + \dots \quad (6.3)$$

For two blocks the mixed multi-index  $e_i + e_j$  contributes

$$\frac{z^{1+\delta_i+\delta_j}}{p^2}(\mathcal{D}+1+\delta_i+\delta_j+p)(B_iB_j) = \frac{z^{1+\delta_i+\delta_j}}{p^2}((1+\delta_i+\delta_j+p)B_iB_j + B'_iB_j + B_iB'_j), \quad (6.4)$$

which shows why the inverse of a sum of perturbations is not the sum of the separately computed inverse corrections.

## 6.3 Degrees and leading logarithms

**Proposition 6.3** (Degrees). *Let  $d_i = \deg B_i$  and  $D(\mathbf{k}) = \sum_i k_i d_i$ . Then  $\deg C_{\mathbf{k}}^{(r)} \leq D(\mathbf{k})$ , with coefficient at the largest possible degree*

$$[L^{D(\mathbf{k})}] C_{\mathbf{k}}^{(r)} = \frac{(-1)^n r}{p^n \mathbf{k}!} \prod_{j=1}^{n-1} (r + \sigma + pj) \prod_i ([L^{d_i}] B_i)^{k_i}, \quad (6.5)$$

which is nonzero exactly when  $r \neq 0$  and  $r + \sigma + pj \neq 0$  for  $1 \leq j \leq n-1$  (always true for  $r > 0$ ,  $p > 0$ ). Under those conditions the degree is exactly  $D(\mathbf{k})$ . Zero operator constants differentiate and may lower the degree or annihilate the polynomial, while  $r = 0$  annihilates every correction. An aggregated block  $C_{\sigma}^{(r)}$  has degree at most  $\max\{D(\mathbf{k}) : \mathbf{k} \cdot \boldsymbol{\delta} = \sigma\}$ ; equality requires a nonzero sum of its top-degree coefficients.

*Proof.* Each factor  $\mathcal{D} + c$  with  $c \neq 0$  preserves the degree and multiplies the top coefficient by  $c$ ; a factor with  $c = 0$  differentiates. The product  $B^{\mathbf{k}}$  has degree  $D(\mathbf{k})$ , and the prefactor contains  $r$ .  $\square$

For the log example ( $a = p = \delta = 1$ ,  $B = 1 + L$ , single index  $n$ ) the formula reads

$$\begin{aligned} P_n(L) &= \frac{(-1)^n}{n!} \prod_{j=n+2}^{2n} (\mathcal{D} + j) (1 + L)^n, \\ [L^n] P_n &= \frac{(-1)^n}{n!} \prod_{j=n+2}^{2n} j = (-1)^n \frac{(2n)!}{n! (n+1)!} = (-1)^n \text{Cat}_n, \end{aligned} \quad (6.6)$$

the signed Catalan numbers. The next coefficient is also explicit: selecting the derivative in exactly one factor of the product gives

$$[L^{n-1}]P_n = (-1)^n \text{Cat}_n n (1 + H_{2n} - H_{n+1}), \quad H_k = \sum_{j \leq k} 1/j, \quad (6.7)$$

because  $\prod_{j=n+2}^{2n} (\mathcal{D} + j)(1 + L)^n$  has  $L^{n-1}$ -coefficient  $(\prod j)(n + \sum_{j=n+2}^{2n} n/j)$  (the term  $n$  from expanding  $(1 + L)^n$  and the terms  $n/j$  from applying  $\mathcal{D}$  in the factor  $j$ ). Every  $P_n$  is divisible by  $1 + L$ : in the variable  $t = 1 + L$  the operator product  $\prod_{j=n+2}^{2n} (\mathcal{D} + j)$  consists of  $n - 1$  factors, each of which either differentiates (lowering the power of  $t$  by one) or multiplies by a constant, so every monomial of  $\prod (\mathcal{D} + j) t^n$  has degree at least  $n - (n - 1) = 1$  in  $t$ ; the package confirms the factorisation through  $n = 8$ . Thus

$$\begin{aligned} P_1 &= -(1 + L), & P_2 &= (1 + L)(2L + 3), & P_3 &= -\frac{1}{2}(1 + L)(10L^2 + 31L + 23), \\ P_4 &= \frac{1}{6}(1 + L)(84L^3 + 398L^2 + 607L + 299), \end{aligned}$$

which are the blocks of (1.3).

#### 6.4 Constant coefficients: Fuss–Catalan numbers

When every  $B_i$  is a constant  $b_i$ , the operators act as scalars and

$$C_{\mathbf{k}}^{(r)} = \frac{(-1)^n r}{p^n \mathbf{k}!} \prod_{j=1}^{n-1} (r + \sigma + pj) \prod_i b_i^{k_i} = \frac{r}{r + \sigma} \binom{-(r + \sigma)/p}{n} \frac{n!}{\mathbf{k}!} \prod_i b_i^{k_i}. \quad (6.8)$$

The binomial expression in (6.8) is read by its removable limit when  $r + \sigma = 0$ ; the operator-product expression has no division by  $r + \sigma$  and remains directly valid. For  $f(x) = x + bx^\alpha$  ( $p = 1$ ,  $\delta = \alpha - 1$ ,  $r = 1$ ) this is the generalised Fuss–Catalan law

$$\begin{aligned} g(y) &= \sum_{n \geq 0} c_n b^n y^{1+n(\alpha-1)}, \\ c_n &= \frac{(-1)^n}{n!} \prod_{j=0}^{n-2} (n\alpha - j) = \frac{(-1)^n}{1 + n(\alpha - 1)} \binom{n\alpha}{n} = (-1)^n \frac{\Gamma(n\alpha + 1)}{\Gamma(n + 1)\Gamma(n(\alpha - 1) + 2)}, \end{aligned} \quad (6.9)$$

valid for every real  $\alpha > 1$  (no rationality is used), with  $c_0 = 1$  understood separately in the product formula. For  $\alpha = 2$  the  $c_n$  are the signed Catalan numbers and, when  $b \neq 0$ ,  $g = (\sqrt{1 + 4by} - 1)/(2b)$ ; the limit at  $b = 0$  is  $g = y$ . For  $b = 1$  and  $\alpha = \sqrt{2}$ ,

$$\begin{aligned} c_1 &= -1, & c_2 &= \sqrt{2}, & c_3 &= -\frac{6 - \sqrt{2}}{2}, \\ c_4 &= \frac{17\sqrt{2}}{3} - 4, & c_5 &= \frac{51\sqrt{2}}{4} - \frac{305}{12}, & c_6 &= \frac{322\sqrt{2}}{5} - 77, \end{aligned}$$

which gives (1.4). In this one-gap case  $g(y) = yV(by^{\alpha-1})$  with  $V$  the solution of  $V + tV^\alpha = 1$ ,  $V(0) = 1$ , analytic at  $t = 0$  with radius of convergence

$$R_\alpha = \frac{(\alpha - 1)^{\alpha-1}}{\alpha^\alpha}, \quad |c_n| \sim \frac{\sqrt{\alpha}}{\sqrt{2\pi}(\alpha - 1)^{3/2}} n^{-3/2} R_\alpha^{-n}, \quad (6.10)$$

by Stirling's formula, so the expansion (1.4) converges (and equals the positive inverse, by analytic continuation of the implicit equation) for  $0 < y < R_{\sqrt{2}}^{1/(\sqrt{2}-1)} \approx 0.12686$ . The singularity at  $t = -R_\alpha$  is the fold  $V = \alpha/(\alpha - 1)$  of the implicit equation on the negative axis.

## 6.5 Formal existence, uniqueness, and the two iterative engines

Theorem 6.1 produces the coefficients; the following independent argument shows that they are the only possible ones and provides two alternative algorithms. Write  $g = z(1 + U)$  with  $U$  an unknown jet of positive valuation. Then  $f(g) = y$  is equivalent to

$$E(U) := (1 + U)^p \left( 1 + \sum_i z^{\delta_i} (1 + U)^{\delta_i} B_i(L + \log(1 + U)) \right) - 1 = 0, \quad (6.11)$$

an equation in the jet algebra (Lemma 2.7 evaluates every term).

**Proposition 6.4** (Triangular structure). *Equation (6.11) has exactly one formal solution  $U$  of positive valuation with support in the semigroup  $S$  generated by the gaps. Its coefficient at  $z^\sigma$  is determined by the coefficients at smaller weights through one division by  $p$ .*

*Proof.*  $E(U) = pU + \sum_i z^{\delta_i} B_i(L) + (\text{terms of weight } \geq \min(2 \text{ val } U, \text{ val } U + \delta_*))$ , because  $(1 + U)^p = 1 + pU + \mathcal{O}(U^2)$  and every correction carries a factor  $z^{\delta_i}$ . Order the elements of  $S$  increasingly (Lemma 2.4); at a new weight  $\sigma$  the coefficient of  $z^\sigma$  in  $E(U)$  is  $pU_\sigma$  plus an expression in the  $U_{\sigma'}$  with  $\sigma' < \sigma$ , which fixes  $U_\sigma$ . If  $U, V$  are two solutions with  $U - V$  of least weight  $\sigma$ , then  $E(U) - E(V) = (U - V)(p + \text{positive weight})$  has a nonzero block at  $z^\sigma$ , a contradiction.  $\square$

**Picard iteration.** Rewriting (6.11) as  $U = \Psi(U) := (1 + \sum_i z^{\delta_i} (1 + U)^{\delta_i} B_i(L + \log(1 + U)))^{-1/p} - 1$ , the map  $\Psi$  gains one gap per step:  $\text{val}(\Psi(U) - \Psi(V)) \geq \text{val}(U - V) + \delta_*$ . Hence  $U_{k+1} = \Psi(U_k)$ ,  $U_0 = 0$ , agrees with the solution below weight  $(k + 1)\delta_*$ , and  $\lceil H/\delta_* \rceil$  iterations give all blocks below  $H$ .

**Newton iteration.** With  $E'(U) = p(1 + U)^{p-1} + \sum_i z^{\delta_i} (1 + U)^{p+\delta_i-1} ((p + \delta_i)B_i + B'_i)(L + \log(1 + U))$ , a unit of the jet algebra with constant leading coefficient  $p$ , the iteration  $U_{k+1} = U_k - E(U_k)/E'(U_k)$  doubles the valuation of the error: if  $\text{val}(U_k - U) \geq s$  then  $\text{val}(U_{k+1} - U) \geq 2s$ , because  $E(U_k) = E'(U)(U_k - U) + \mathcal{O}((U_k - U)^2)$  and  $E'(U_k)^{-1} = E'(U)^{-1}(1 + \mathcal{O}(U_k - U))$ . Starting from  $U_0 = 0$  with  $s = \delta_*$ , about  $\log_2(H/\delta_*)$  iterations suffice. The package offers this engine as `Method -> "Newton"`; it and the multi-index formula must agree, which is a strong internal check.

## 7 Convergence

A formal identity such as (6.1) says nothing, by itself, about the function  $g$ : the series might diverge, or converge to something else. This section proves that the expansion converges absolutely for small  $z$  and equals the branch of Theorem 4.1, first by an implicit-function argument that also yields the coefficient bounds needed for the remainder theory, then by a Rouché argument that yields explicit constants.

### 7.1 Convergence in finitely many small parameters

**Theorem 7.1** (Convergence). *Let  $f$  be the germ (4.1) and  $g$  its branch. For every real  $r$  the series (6.1) converges absolutely for all sufficiently small  $z > 0$ , its sum is  $g(y)^r$ , and there are constants  $M, K$  such that*

$$|C_{\mathbf{k}}^{(r)}(L)| \leq M K^{|\mathbf{k}|} M(z)^{D(\mathbf{k})} \quad (L \leq \log z_1), \quad D(\mathbf{k}) = \sum_i k_i \deg B_i. \quad (7.1)$$

*In particular  $g^r \in \mathcal{E}$  (Definition 3.1) with support  $r + S$ .*

*Proof.* Write  $g = zw$  with  $w \rightarrow 1$ ; by (6.11) the unknown  $w$  solves

$$w^p \left( 1 + \sum_i z^{\delta_i} w^{\delta_i} B_i(L + \log w) \right) = 1.$$

Expand each  $B_i(L + \log w) = \sum_{j=0}^{d_i} L^j A_{ij}(\log w)$  with  $A_{ij}(v) = \sum_{s \geq j} b_{is} \binom{s}{j} v^{s-j}$ , where  $B_i = \sum_s b_{is} L^s$ . Introduce independent complex parameters  $T_{ij}$  in place of  $z^{\delta_i} L^j$  and consider

$$\mathcal{E}(w, T) = w^p \left( 1 + \sum_{i,j} T_{ij} w^{\delta_i} A_{ij}(\log w) \right) - 1. \quad (7.2)$$

On the disc  $|w - 1| < 1$  with the principal logarithm,  $\mathcal{E}$  is analytic in  $(w, T)$ ;  $\mathcal{E}(1, 0) = 0$  and  $\partial_w \mathcal{E}(1, 0) = p \neq 0$ . The analytic implicit function theorem gives an analytic  $W(T)$  on a polydisc  $|T_{ij}| < \rho$  with  $W(0) = 1$  and  $\mathcal{E}(W(T), T) = 0$ , and  $W^r$  is analytic there too. Its Taylor series  $W(T)^r = \sum_{\nu} c_{\nu} T^{\nu}$  converges absolutely on the smaller polydisc  $|T_{ij}| \leq \rho/2$ , with the Cauchy estimates  $|c_{\nu}| \leq M(2/\rho)^{|\nu|}$ .

Now specialise  $T_{ij} = z^{\delta_i} L^j$ . Since  $z^{\delta_i} M(z)^{d_i} \rightarrow 0$ , the point lies in the polydisc for small  $z$ , and  $W(T(z))$  is real (all data are real and the solution is unique) and close to 1, so  $zW(T(z))$  is a solution of  $f(u) = y$  with  $u \sim z$ ; by Theorem 4.1 it is  $g(y)$ . Group the absolutely convergent series  $\sum_{\nu} c_{\nu} \prod_{i,j} (z^{\delta_i} L^j)^{\nu_{ij}}$  by  $k_i = \sum_j \nu_{ij}$ . There are at most  $\prod_i (d_i + 1)^{k_i}$  terms for a given  $\mathbf{k}$ , with total logarithmic degree  $\sum_{i,j} j \nu_{ij} \leq D(\mathbf{k})$ . Their sum is  $z^{\mathbf{k} \cdot \delta} \tilde{C}_{\mathbf{k}}(L)$ , and the Cauchy bounds give both (7.1) and  $\|\tilde{C}_{\mathbf{k}}\|_1 \leq MK^{|\mathbf{k}|}$  for  $K = (2/\rho) \max_i (d_i + 1)$ . To identify the individual multi-index coefficients even when weights coincide, retain independent markers  $\eta_i$  by substituting  $T_{ij} = \eta_i z^{\delta_i} L^j$ . Both constructions solve the same formal implicit equation in the markers, whose linearization in  $w$  is  $p \neq 0$ , so its solution is unique coefficientwise. Thus  $\tilde{C}_{\mathbf{k}} = C_{\mathbf{k}}^{(r)}$ . Finally, aggregation by equal weights preserves the bounds of Definition 3.1: all representations of weight  $\sigma$  have length between  $\sigma/\delta_{\max}$  and  $\sigma/\delta_*$ , so the shortest representation controls their lengths up to a fixed factor; their count is at most exponential in that length.  $\square$

*Remark 7.2.* The theorem gives convergence of a multivariate power series after the substitution  $T_{ij} = z^{\delta_i} (\log z)^j$ . The normalized series  $g^r/z^r$  converges uniformly after shrinking  $z_1$  so all these parameters remain in a smaller polydisc; for  $r < 0$  this does not assert absolute uniform convergence of  $g^r$  itself on  $(0, z_1)$ . A single-variable radius exists in special cases such as (6.10). Nothing is uniform in  $a, p, \delta_i, B_i$ : for  $x + x^\alpha$  as  $\alpha \downarrow 1$ , the radius  $R_\alpha$  in  $t = y^{\alpha-1}$  tends to 1, whereas the corresponding endpoint  $R_\alpha^{1/(\alpha-1)}$  in  $y$  tends to 0.

## 7.2 An explicit conditional majorant

The implicit-function argument has no explicit constants. Rouché's theorem in the marker equation gives them, at the price of a hypothesis that must be checked at each  $y$ .

**Theorem 7.3** (Explicit tail bound). *Consider  $f(t) = t + h(t)$ ,  $h(t) = \sum_i t^{1+\alpha_i} P_i(\log t)$  with real  $\alpha_i$  and  $P_i(L) = \sum_s p_{is} L^s$ . The general germ reduces to this form by  $t = u^p$ ,  $q = v/a$ , with  $\alpha_i = \delta_i/p$ ; these are positive at  $t \rightarrow 0$  when  $p > 0$  and negative at  $t \rightarrow \infty$  when  $p < 0$ . Fix  $0 < r < 1$ , put  $c_r = -\log(1 - r)$ ,*

$$A_r(b) = \max\{(1 - r)^b, (1 + r)^b\},$$

and

$$\mathcal{Q}_r(t) = \frac{1}{r} \sum_i t^{\alpha_i} A_r(1 + \alpha_i) \sum_s |p_{is}| (|\log t| + c_r)^s. \quad (7.3)$$

Let  $t > 0$  satisfy  $\mathcal{Q}_r(t) < 1$ . Then:

1. for every complex  $\lambda$  with  $|\lambda| < 1/\mathcal{Q}_r(t)$  the equation  $u + \lambda h(u) = t$  has exactly one solution  $u(\lambda)$  in the disc  $|u - t| < rt$ ; it is analytic in  $\lambda$ ,  $u(0) = t$ , and  $u(\lambda)$  is real for real  $\lambda$ ;
2. the Lagrange terms  $u_N = \frac{(-1)^N}{N!} \frac{d^{N-1}}{dt^{N-1}} [h(t)^N]$  satisfy  $|u_N| \leq \frac{rt}{N} \mathcal{Q}_r(t)^N$ , and  $u(\lambda) = t + \sum_{N \geq 1} u_N \lambda^N$  converges absolutely for  $|\lambda| < 1/\mathcal{Q}_r(t)$ , in particular at  $\lambda = 1$ ;
3.  $u(1)$  is the positive solution of  $f(u) = t$  in  $(t(1-r), t(1+r))$ . It coincides with any local branch  $g(t) \sim t$  near the endpoint under consideration, once  $t$  is sufficiently close to that endpoint, and

$$\left| u(1) - t - \sum_{N=1}^{N_0} u_N \right| \leq \frac{rt}{N_0 + 1} \frac{\mathcal{Q}_r(t)^{N_0+1}}{1 - \mathcal{Q}_r(t)}. \quad (7.4)$$

*Proof.* On the circle  $|s - t| = rt$  write  $s = t(1 + \zeta)$  with  $|\zeta| = r < 1$ . For a real exponent  $b$ ,  $|s^b| = |s|^b \leq t^b A_r(b)$ ; using only the upper bound on  $|s|$  would be incorrect for  $b < 0$ . The principal logarithm satisfies  $|\log s| \leq |\log t| + |\log(1 + \zeta)| \leq |\log t| + c_r$ . Hence  $|h(s)| \leq rt \mathcal{Q}_r(t)$ . Rouché's theorem gives exactly one zero, counted with multiplicity, whenever  $|\lambda| \mathcal{Q}_r(t) < 1$ . That zero is simple, analytic in  $\lambda$ , and real for real  $\lambda$  by conjugation and uniqueness. Cauchy's estimate in the marker formula gives

$$|u_N| \leq \frac{1}{N!} \frac{(N-1)! (rt \mathcal{Q}_r)^N}{(rt)^{N-1}} = \frac{rt}{N} \mathcal{Q}_r^N.$$

The root at  $\lambda = 1$  is positive and lies in the indicated interval. Any branch with  $g(t) \sim t$  eventually lies in that interval too, so the roots agree. Finally sum the bound over  $N > N_0$ , using  $1/N \leq 1/(N_0 + 1)$ .  $\square$

**Corollary 7.4** (Exponent cutoff). *Assume  $\alpha_i > 0$  and  $B \geq 0$ . Let  $G_B$  retain all multi-indices of weight  $\mathbf{k} \cdot \boldsymbol{\alpha} \leq B$ . Every multi-index with  $|\mathbf{k}| \leq N_0 := \lfloor B/\alpha_{\max} \rfloor$  is retained, so under  $\mathcal{Q}_r(t) < 1$*

$$|g(t) - G_B(t)| \leq \frac{rt}{N_0 + 1} \frac{\mathcal{Q}_r(t)^{N_0+1}}{1 - \mathcal{Q}_r(t)}.$$

*Proof.* The omitted part of the multi-marker expansion is a subseries of  $\sum_{N > N_0} u_N$  (the expansion of  $u_N$  into multi-indices is absolutely convergent by the same Cauchy estimates, since  $\sum_{|\mathbf{k}|=N} \frac{\prod M_i^{k_i}}{\mathbf{k}!} = \frac{(\sum M_i)^N}{N!}$ ).  $\square$

**Example 7.5.** For  $f_1(x) = x + x^2(1 + \log x)$  and  $r = \frac{1}{2}$ :  $\mathcal{Q}_{1/2}(y) = \frac{9}{2} y (1 + |\log y| + \log 2)$ , so for  $y \leq \frac{1}{100}$  (where  $\mathcal{Q}_{1/2} \leq 0.284$ ) the expansion converges and the error after  $N_0$  blocks is at most  $\frac{y}{2(N_0+1)} \mathcal{Q}_{1/2}(y)^{N_0+1} / (1 - \mathcal{Q}_{1/2}(y))$ ; at  $y = 10^{-2}$  and  $N_0 = 3$  this is  $1.1 \cdot 10^{-5}$  while the actual error is  $1.5 \cdot 10^{-7}$ . The bound is conditional (it says nothing when  $\mathcal{Q}_r \geq 1$ ) and not tight, but it is rigorous, cheap, and turns the asymptotic statements into inequalities at specific points.

## 8 Remainders after a cutoff

### 8.1 Exponent cutoff: the first omitted block

Let the target cutoff be an exponent  $q$  of the local variable of the result, exclusive. In terms of weights, with  $r$  the observable power, a block  $z^{r+\sigma} C(L)$  has exponent  $(r + \sigma)/|p|$  in the local variable of  $y$  (Section 10), so the retained set is

$$I_H = \{\mathbf{k} \in \mathbb{N}^m : \mathbf{k} \cdot \boldsymbol{\delta} < H\}, \quad H = |p| q - r, \quad (8.1)$$

a finite downward-closed set (Lemma 2.4). Set  $C_0^{(r)} = 1$ . When  $H > 0$  its *one-step boundary* is

$$\partial I_H = \{\mathbf{k} + e_i : \mathbf{k} \in I_H, 1 \leq i \leq m\} \setminus I_H, \quad (8.2)$$

also finite. When  $H \leq 0$ ,  $I_H$  is empty and we define its boundary to be  $\{0\}$ , since the leading monomial is already omitted. With at least one correction gap, put

$$\sigma_* = \min\{\boldsymbol{\nu} \cdot \boldsymbol{\delta} : \boldsymbol{\nu} \in \partial I_H\}, \quad C_{\sigma_*}^{(r)} = \sum_{\mathbf{k} \cdot \boldsymbol{\delta} = \sigma_*} C_{\mathbf{k}}^{(r)}, \quad k_* = \deg C_{\sigma_*}^{(r)} \quad (k_* = 0 \text{ if } C_{\sigma_*}^{(r)} = 0). \quad (8.3)$$

**Lemma 8.1.**  $\sigma_* = \min(S \setminus [0, H])$ , and every multi-index of weight  $\sigma_*$  lies in  $\partial I_H$ . Thus the complete coefficient at the first omitted weight is obtained from the boundary alone.

*Proof.* For  $H \leq 0$  the assertion is immediate, with  $\sigma_* = 0$ . For  $H > 0$ , let  $\boldsymbol{\nu} \notin I_H$  have weight  $\sigma$ . Along a path from 0 to  $\boldsymbol{\nu}$  that increases one coordinate at a time, the weights increase strictly, and the first point outside  $I_H$  is a boundary point of weight  $\leq \sigma$ ; hence  $\sigma_* \leq \sigma$ , so  $\sigma_*$  is the least weight outside  $[0, H]$ . If  $\sigma = \sigma_*$  the first point outside  $I_H$  has weight exactly  $\sigma_*$  by minimality, and no later point on the path can have the same weight, so  $\boldsymbol{\nu}$  is that boundary point.  $\square$

**Theorem 8.2** (Remainder after an exponent cutoff). Let  $G_H^{(r)} = z^r \sum_{\mathbf{k} \in I_H} z^{\mathbf{k} \cdot \boldsymbol{\delta}} C_{\mathbf{k}}^{(r)}(L)$ . Then, as  $z \rightarrow 0^+$ ,

$$g(y)^r - G_H^{(r)}(y) = z^{r+\sigma_*} C_{\sigma_*}^{(r)}(L) + o(z^{r+\sigma_*}) = \mathcal{O}(z^{r+\sigma_*} M(z)^{k_*}). \quad (8.4)$$

If  $C_{\sigma_*}^{(r)} \neq 0$  the bound is attained by its top logarithm; if  $C_{\sigma_*}^{(r)} = 0$  (a resonant cancellation) the remainder is  $o(z^{r+\sigma_*})$ , and the true order is that of the next nonzero block, which the theorem does not identify.

*Proof.* By Theorem 7.1,  $g^r \in \mathcal{E}$  with the bound (7.1), and the omitted multi-indices are those of weight  $\geq \sigma_*$  by Lemma 8.1; Lemma 3.2 (refined form) gives the claim.  $\square$

**Corollary 8.3** (In the variable of the result). With  $w$  the local variable of  $y$  ( $w = |y - y_0|$  for finite  $y_0$ ,  $w = 1/|y|$  for infinite  $y_0$ ),

$$g(y)^r - G_H^{(r)}(y) = \mathcal{O}(w^{(r+\sigma_*)/|p|} M(w)^{k_*}), \quad \frac{r + \sigma_*}{|p|} \geq q,$$

since  $z = (w/|a|)^{1/p}$  for  $p > 0$ ,  $z = (|a|w)^{1/|p|}$  for  $p < 0$ , and  $M(z) \asymp M(w)$ .

For a pure monomial ( $m = 0$ ), the inverse observable is exactly  $z^r$ : the tail is zero if this term is retained and equals  $z^r$  otherwise. The package reports the pair  $((r + \sigma_*)/|p|, k_*)$  as `PowerLogRemainder[w,  $\beta$ , k]` and the block  $z^{r+\sigma_*} C_{\sigma_*}^{(r)}$  as `"FrontierTerm"`. If that coefficient vanishes, a sharper order requires computing further weights or proving that the remaining series terminates.

**Example 8.4** (Resonances). For  $f = x + x^2 + x^3$  the gaps are 1 and 2; at weight 2 the indices (2, 0) and (0, 1) contribute 2 and  $-1$ , at weight 3 the indices (3, 0) and (1, 1) contribute  $-5$  and  $+5$ , so

$$g(y) = y - y^2 + y^3 + 0 \cdot y^4 - 4y^5 + 14y^6 - 30y^7 + 33y^8 + \dots,$$

and after the cutoff 4 the reported remainder is  $\mathcal{O}(y^4)$  while the truth is  $\mathcal{O}(y^5)$ . For  $f = x + x^2 + 2x^3$  the block at  $y^3$  cancels instead. Exact arithmetic on the exponents is what makes such cancellations detectable at all: with floating-point exponents, (2, 0) and (0, 1) might not be recognised as the same weight.



**Example 8.5** (The log example). Here  $m = 1$ ,  $\delta = 1$ ,  $B = 1 + L$ , so  $I_H = \{0, \dots, N\}$  for  $H = N + 1$ , the boundary is  $\{N + 1\}$ , and by (6.6) the frontier block  $y^{N+2}P_{N+1}(\log y)$  has degree exactly  $N + 1$  with top coefficient  $(-1)^{N+1} \text{Cat}_{N+1}$ . Hence

$$g_1(y) = y + \sum_{n=1}^N y^{n+1} P_n(\log y) + \mathcal{O}(y^{N+2} M(y)^{N+1}),$$

$$\frac{g_1(y) - y - \sum_{n \leq N} y^{n+1} P_n(\log y)}{y^{N+2} (\log y)^{N+1}} \rightarrow (-1)^{N+1} \text{Cat}_{N+1}, \quad (8.5)$$

and in particular after  $y - y^2(1 + \log y)$  the error is  $y^3(2 \log^2 y + 5 \log y + 3) + \dots$ , not  $\mathcal{O}(y^3)$ . The ratio in (8.5) converges slowly (like  $1 + c/\log y$ ): for  $N = 1$  it is 1.89 at  $y = 10^{-20}$  and 1.98 at  $y = 10^{-100}$ .

## 8.2 Depth truncation

When the exponents are symbolic (Section 12) their order may be undecidable, but the total degree  $|\mathbf{k}|$  of a multi-index is always known.

**Theorem 8.6** (Remainder after a depth cutoff). *Let  $G_{\leq N}^{(r)} = z^r \sum_{|\mathbf{k}| \leq N} z^{\mathbf{k} \cdot \delta} C_{\mathbf{k}}^{(r)}(L)$ ,  $\delta_* = \min_i \delta_i$  and  $d_* = \max_i \deg B_i$ . Then*

$$g(y)^r - G_{\leq N}^{(r)}(y) = \mathcal{O}(z^{r+(N+1)\delta_*} M(z)^{(N+1)d_*}). \quad (8.6)$$

*Proof.* The omitted terms have  $|\mathbf{k}| \geq N + 1$ ; use (7.1) and the count  $m^n$  of multi-indices of degree  $n$  exactly as in the proof of Lemma 3.2.  $\square$

Depth truncation retains every block of exponent below  $r + (N + 1)\delta_*$ , but it may also retain blocks above that bound while omitting smaller ones when several gaps are present: for  $f = x + x^2 + x^4$  depth 1 gives  $y - y^2 - y^4$  although the omitted depth-two term  $2y^3$  dominates  $y^4$ . The remainder (8.6) is nevertheless correct as stated.

## 8.3 Numerical illustration

Table 1 shows, for the two examples of [1] and for the inverse of (1.5) at infinity, the actual error  $|g(y) - G(y)|$  of the truncated expansion (the exact inverse computed by `FindRoot` at 80 digits) and its ratio to the reported remainder scale  $w^{\beta_*} M(w)^{k_*}$ . The ratios stay bounded, and in the log example they grow slowly with  $N$  because the leading coefficient of the frontier block,  $\text{Cat}_{N+1}$ , does.

## 9 Transport of remainders and finite smoothness

The theory so far concerns an exact finite germ. In practice the forward function is often known only through a truncated expansion, and one wants to know how the omitted part affects the inverse; conversely, one wants to turn a small residual  $f(G(y)) - y$  of an approximation  $G$  into a bound for  $G - g$ . Both are mean-value arguments, but the scaling factor  $z^{1-p}$  of the derivative must not be forgotten.

| $f_1 = x + x^2(1 + \log x)$ , $N$ corrections               | $y = 10^{-2}$       |       | $y = 10^{-4}$        |       | $y = 10^{-8}$        |       |
|-------------------------------------------------------------|---------------------|-------|----------------------|-------|----------------------|-------|
|                                                             | error               | ratio | error                | ratio | error                | ratio |
| $N = 1$                                                     | $2.4 \cdot 10^{-5}$ | 0.77  | $1.3 \cdot 10^{-10}$ | 1.22  | $5.9 \cdot 10^{-22}$ | 1.56  |
| $N = 2$                                                     | $1.8 \cdot 10^{-6}$ | 1.03  | $2.4 \cdot 10^{-13}$ | 2.26  | $2.5 \cdot 10^{-28}$ | 3.38  |
| $N = 3$                                                     | $1.5 \cdot 10^{-7}$ | 1.51  | $5.1 \cdot 10^{-16}$ | 4.69  | $1.2 \cdot 10^{-34}$ | 8.18  |
| $N = 5$                                                     | $1.2 \cdot 10^{-9}$ | 3.75  | $2.7 \cdot 10^{-21}$ | 24.0  | $3.1 \cdot 10^{-47}$ | 57.4  |
| <hr/>                                                       |                     |       |                      |       |                      |       |
| $f_2 = x + x^{\sqrt{2}}$ , $N$ corrections                  |                     |       |                      |       |                      |       |
| $N = 1$                                                     | $2.5 \cdot 10^{-4}$ | 1.14  | $6.6 \cdot 10^{-8}$  | 1.37  | $3.3 \cdot 10^{-15}$ | 1.41  |
| $N = 2$                                                     | $6.0 \cdot 10^{-5}$ | 1.82  | $2.4 \cdot 10^{-9}$  | 2.21  | $2.6 \cdot 10^{-18}$ | 2.29  |
| $N = 3$                                                     | $1.5 \cdot 10^{-5}$ | 3.16  | $9.1 \cdot 10^{-11}$ | 3.86  | $2.2 \cdot 10^{-21}$ | 4.01  |
| $N = 5$                                                     | $1.2 \cdot 10^{-6}$ | 10.9  | $1.5 \cdot 10^{-13}$ | 13.5  | $1.8 \cdot 10^{-27}$ | 14.1  |
| <hr/>                                                       |                     |       |                      |       |                      |       |
| <hr/>                                                       |                     |       |                      |       |                      |       |
| inverse of $(1 + x + x^{\sqrt{2}})^{\sqrt{2}}$ , $n$ blocks |                     |       |                      |       |                      |       |
|                                                             | $y = 10^4$          |       | $y = 10^8$           |       | $y = 10^{16}$        |       |
| $n = 2$                                                     | 0.72                | 0.33  | 1.9                  | 0.39  | 9.3                  | 0.40  |
| $n = 4$                                                     | $9.6 \cdot 10^{-2}$ | 0.65  | $1.5 \cdot 10^{-2}$  | 0.69  | $3.4 \cdot 10^{-4}$  | 0.71  |
| $n = 7$                                                     | $2.8 \cdot 10^{-4}$ | 0.039 | $1.2 \cdot 10^{-6}$  | 0.023 | $4.0 \cdot 10^{-11}$ | 0.015 |

Table 1: Recorded errors of truncated expansions and their ratio to the remainder scale. In the third group the local variable is  $1/y$  and the leading inverse term is  $\sqrt{y}$ ; for the two-block approximation the absolute error grows while its relative error decreases.

### 9.1 From the forward function to the inverse

**Theorem 9.1** (Transport of a forward remainder). *Let  $f$  be the germ (4.1) with  $p \neq 0$  and let  $F = f + R$  on  $(0, u_0)$ , where*

$$R(u) = \mathcal{O}(u^\rho M(u)^k), \quad R'(u) = \mathcal{O}(u^{\rho-1} M(u)^k), \quad \rho > p, \quad k \in \mathbb{N}. \quad (9.1)$$

*Then  $F$  has a local inverse  $G$  on the same branch as  $g$ , and*

$$G(y) - g(y) = \mathcal{O}(z^{\rho-p+1} M(z)^k) = \mathcal{O}(w^{(\rho-p+1)/|p|} M(w)^k), \quad (9.2)$$

*where  $w = |y - y_0|$  for  $p > 0$  and  $w = 1/|y|$  for  $p < 0$ . For any fixed real observable power  $r$ ,*

$$G(y)^r - g(y)^r = \mathcal{O}(z^{\rho-p+r} M(z)^k) = \mathcal{O}(w^{(\rho-p+r)/|p|} M(w)^k). \quad (9.3)$$

*For  $r = 0$  this difference is identically zero. The same conclusions hold if the derivative hypothesis is replaced by the assumption that  $F$  is continuous and strictly monotone near 0.*

*Proof.* From (9.1) and Lemma 2.2,  $F(u) - y_0 \sim au^p$  and  $F'(u) \sim apu^{p-1}$ , so  $F$  is strictly monotone near 0 and its inverse  $G$  satisfies  $G \sim z$  (Theorem 4.1 applies verbatim). Put  $u = G(y)$ ,  $v = g(y)$ ; both are  $\sim z$ . Then  $F(v) - y = R(v) = \mathcal{O}(z^\rho M(z)^k)$ , while  $F(u) - y = 0$ . On the interval between  $u$  and  $v$ , which lies in  $[z/2, 2z]$  for small  $z$ ,  $F'$  is at least  $\frac{1}{2}|ap|z^{p-1}2^{-|p-1|}$  in absolute value. The mean value theorem gives (9.2). With continuous strict monotonicity instead of the derivative bound, the leading equivalence still supplies a continuous local inverse with  $G \sim z$ , and  $f(u) - f(v) = -R(u)$  together with  $|f'| \asymp z^{p-1}$  gives the same estimate. Finally the derivative of  $s^r$  on  $[z/2, 2z]$  is  $\mathcal{O}(z^{r-1})$ , giving (9.3).  $\square$

**Remark 9.2.** The derivative hypothesis is not a consequence of the value bound:  $R(u) = u^2 \sin(u^{-2})$  is  $\mathcal{O}(u^2)$ , but  $R'(u) = 2u \sin(u^{-2}) - 2u^{-1} \cos(u^{-2})$  destroys the monotonicity of  $u + R(u)$ . Nor does non-strict monotonicity alone ensure an inverse, since it permits plateaus. Expansions obtained by the convergent structural operations above admit corresponding derivative estimates, but a formal `Series` result is not itself a proof of them. A declared `"InputRemainder"`  $\rightarrow \{\rho, k\}$  carries the derivative or continuous strict monotonicity hypothesis of this theorem.

**Corollary 9.3** (Cutoff cap). *The data justify every block of  $G^r$  strictly below the exponent  $(\rho - p + r)/|p|$  in  $w$ . Thus a requested cutoff  $q$  is certified from those data when  $q \leq (\rho - p + r)/|p|$ ; the reported remainder is the coarser of the truncation remainder and (9.3), compared by exponent first and by logarithmic degree at equal exponents. A larger requested cutoff requires a sharper forward remainder or additional information. For powers of the original variable at an infinite source endpoint, replace  $r$  here by the local-coordinate power  $-r$ .*

For example  $\sin x + x^2 \log x = x + x^2 \log x - x^3/6 + \mathcal{O}(x^5)$  is a germ with  $p = 1$ , gaps 1, 2 and  $\rho = 5$ ,  $k = 0$ ; the inverse can be requested up to the cutoff  $q = 5$  and is  $y - y^2 \log y + y^3(\frac{1}{6} + \log y + 2 \log^2 y) + \mathcal{O}(y^4 M^3)$  for  $q = 4$ .

## 9.2 From a residual to the error of the inverse

**Theorem 9.4** (Residual bracket). *Let  $F$  be continuous on  $[A, B]$  and differentiable inside with  $F' \geq \mu > 0$ . Let  $x_0 \in (A, B)$ ,  $y \in \mathbb{R}$ ,  $\varrho = F(x_0) - y$ . If  $[x_0 - |\varrho|/\mu, x_0 + |\varrho|/\mu] \subseteq [A, B]$ , then  $F(x) = y$  has exactly one solution  $x_*$  in  $[A, B]$ , it lies in that interval, and*

$$|x_* - x_0| \leq \frac{|\varrho|}{\mu}. \quad (9.4)$$

*If also  $F' \leq M_1 < \infty$  on  $(A, B)$ , then  $|x_* - x_0| \geq |\varrho|/M_1$ .*

*Proof.*  $F(x_0 + |\varrho|/\mu) \geq F(x_0) + |\varrho| \geq y$  and  $F(x_0 - |\varrho|/\mu) \leq y$ ; the intermediate value theorem gives a root in the interval, monotonicity its uniqueness in  $[A, B]$ , and the mean value theorem gives the error bounds. A finite derivative maximum was not part of the hypotheses, so the lower error bound is stated separately.  $\square$

This gives an a posteriori certificate that needs no expansion theory, only a lower bound on the derivative. For the two examples the global bounds (4.5) give, for every  $y > 0$  and every  $G > 0$ ,

$$|G - g_1(y)| \leq \frac{|G + G^2(1 + \log G) - y|}{1 - 2e^{-5/2}}, \quad |G - g_2(y)| \leq |G + G^{\sqrt{2}} - y|. \quad (9.5)$$

For a general germ,  $|f'| \asymp z^{p-1}$ . For an approximation  $G \sim z$ , apply the bracket on  $[z/2, 2z]$  with  $\mu = \frac{1}{2}|ap|z^{p-1}2^{-|p-1|}$ , after replacing  $(f, y)$  by  $(-f, -y)$  if the derivative is negative. A residual small enough that the bracket lies in this interval yields  $|G - g| \leq \frac{2^{1+|p-1|}}{|ap|} z^{1-p} |f(G) - y|$ .

**The residual of the truncated expansion.** Combining Theorem 8.2 with the mean value theorem, the truncated inverse  $G_H$  satisfies

$$\frac{f(G_H(y)) - y_0}{a z^p} - 1 = -p z^{\sigma_*} C_{\sigma_*}(L) + o(z^{\sigma_*}), \quad (9.6)$$

Here  $y_0 = 0$  for  $p < 0$ , and the truncation retains the leading term so  $G_H \sim z$ . Indeed  $f'(s) = apz^{p-1}(1 + \mathcal{O}(z^{\delta_*} M^D))$  between  $g$  and  $G_H$ , with a positive power margin that absorbs the logarithms in the error. Multiplying by  $G_H - g$  proves the stated little- $o$  term. The normalized residual has the first error block multiplied by  $-p$ ; **InverseResidual** computes it in the jet algebra. If the inverse is retained to cutoff  $q$  in  $w$ , its unnormalized forward residual cutoff is  $q + (p - 1)/|p|$ ; the normalized cutoff subtracts  $p/|p|$  from that value.

### 9.3 Finite smoothness suffices

The finite germ is analytic away from 0, and the convergence proof used that. For perturbations outside the finite power-log class (a coefficient  $\sin(\log x)$ , a term  $e^{-1/x}$ , a function known only through derivative bounds) the marker formula (5.3) can still be applied, and the following theorem, which appears in report 05 with a sketched proof, shows that it is a genuine asymptotic expansion whenever the perturbation and finitely many of its derivatives are small in the power-log scale. We give the complete proof.

**Theorem 9.5** (Tame near-identity inversion). *Let  $N \geq 0$ ,  $\delta > 0$  and  $M \geq 0$ . Let  $\varphi$  be real-valued and  $C^{N+1}$  on  $(0, x_0)$  with*

$$\varphi^{(j)}(x) = \mathcal{O}(x^{1+\delta-j}M(x)^M) \quad (x \rightarrow 0^+), \quad j = 0, 1, \dots, N+1. \quad (9.7)$$

*Then  $f = x + \varphi$  has a unique local inverse  $g$  on  $(0, y_0)$  with  $g(y) \sim y$ , and*

$$g(y) = y + \sum_{n=1}^N \frac{(-1)^n}{n!} \frac{d^{n-1}}{dy^{n-1}} [\varphi(y)^n] + \mathcal{O}(y^{1+(N+1)\delta}M(y)^{(N+1)M}). \quad (9.8)$$

*Proof. Step 1: the homotopy.* For  $\lambda \in [0, 1]$  put  $F_\lambda(x) = x + \lambda\varphi(x)$ . By (9.7) with  $j = 1$ ,  $\varphi'(x) \rightarrow 0$ , so there is  $x_1$  with  $F'_\lambda \geq \frac{1}{2}$  on  $(0, x_1)$  for all  $\lambda \in [0, 1]$ ; each  $F_\lambda$  is an increasing bijection of  $(0, x_1)$  onto an interval containing  $(0, y_1)$  for some  $y_1$  independent of  $\lambda$ . Let  $G(\lambda, y)$  be its inverse, so that

$$G + \lambda\varphi(G) = y, \quad G(0, y) = y. \quad (9.9)$$

Since  $|\varphi(x)| \leq x/2$  for  $x < x_1$  (after shrinking  $x_1$ ),  $F_\lambda(y/2) < y < F_\lambda(2y)$  and therefore  $G(\lambda, y) \in [y/2, 2y]$  for all  $\lambda$  and all small  $y$ . The map  $(\lambda, x) \mapsto x + \lambda\varphi(x) - y$  is  $C^{N+1}$  with  $x$ -derivative  $\geq \frac{1}{2}$ , so by the implicit function theorem  $G$  is  $C^{N+1}$  in  $(\lambda, y)$ , with

$$\partial_y G = \frac{1}{1 + \lambda\varphi'(G)} \in [\frac{2}{3}, 2], \quad \partial_\lambda G = -\frac{\varphi(G)}{1 + \lambda\varphi'(G)} = \psi(G) \partial_y G, \quad \psi := -\varphi. \quad (9.10)$$

*Step 2: the transport identity.* For every  $C^n$  function  $\Psi$  and  $1 \leq n \leq N+1$ ,

$$\partial_\lambda^n [\Psi(G)] = \partial_y^{n-1} [\psi(G)^n \partial_y \Psi(G)]. \quad (9.11)$$

For  $n = 1$ :  $\partial_\lambda \Psi(G) = \Psi'(G) \partial_\lambda G = \Psi'(G) \psi(G) \partial_y G = \psi(G) \partial_y [\Psi(G)]$ . Assume (9.11) for  $n$  and all  $\Psi$ . Then, using it once more for the functions  $\psi^n$  and  $\Psi$  in the second step,

$$\begin{aligned} \partial_\lambda^{n+1} [\Psi(G)] &= \partial_y^{n-1} \partial_\lambda [\psi(G)^n \partial_y \Psi(G)] \\ &= \partial_y^{n-1} [\psi(G) \partial_y (\psi(G)^n \partial_y \Psi(G)) \\ &\quad + \psi(G)^n \partial_y (\psi(G) \partial_y \Psi(G))] \\ &= \partial_y^{n-1} \partial_y [\psi(G)^{n+1} \partial_y \Psi(G)], \end{aligned}$$

where the last equality is the product rule read backwards:  $\partial_y [\psi^{n+1} \Psi_y] = (n+1) \psi^n \psi_y \Psi_y + \psi^{n+1} \Psi_{yy}$  and the bracket equals  $n \psi^n \psi_y \Psi_y + \psi^n (\psi_y \Psi_y + \psi \Psi_{yy})$ , the same thing. (Here  $\psi_y$  stands for  $\partial_y [\psi(G)]$ .)

*Step 3: the coefficients.* At  $\lambda = 0$ ,  $G = y$ , so (9.11) with  $\Psi = \text{id}$  gives  $\partial_\lambda^n G(0, y) = \partial_y^{n-1} [\psi(y)^n] = (-1)^n \partial_y^{n-1} [\varphi(y)^n]$ , the Lagrange terms. Taylor's formula in  $\lambda$  with integral remainder gives

$$g(y) = G(1, y) = y + \sum_{n=1}^N \frac{(-1)^n}{n!} \partial_y^{n-1} [\varphi(y)^n] + \frac{1}{N!} \int_0^1 (1-\lambda)^N \partial_\lambda^{N+1} G(\lambda, y) d\lambda.$$

*Step 4: the remainder.* Put  $\kappa(y) = y^\delta M(y)^M$ . Since  $G \in [y/2, 2y]$  and  $M(G) \leq 2M(y)$ , hypothesis (9.7) gives  $\varphi^{(j)}(G) = \mathcal{O}(\kappa y^{1-j})$  uniformly in  $\lambda$ , for  $j \leq N+1$ . We claim that for  $1 \leq j \leq N+1$ ,

$$\partial_y^j G = \mathcal{O}(y^{1-j}) \quad (j \geq 1), \quad \partial_y^j [\varphi(G)] = \mathcal{O}(\kappa y^{1-j}) \quad (j \geq 0), \quad (9.12)$$

uniformly in  $\lambda \in [0, 1]$ . Differentiating (9.9)  $j$  times in  $y$  and solving for  $\partial_y^j G$ :  $(1 + \lambda \varphi'(G)) \partial_y^j G$  equals  $\delta_{j1}$  minus  $\lambda$  times a sum of terms  $\varphi^{(m)}(G) \prod \partial_y^{i_s} G$  with  $2 \leq m$ ,  $\sum i_s = j$ ,  $i_s \geq 1$  (Faà di Bruno). By induction on  $j$  each such term is  $\mathcal{O}(\kappa y^{1-m}) \prod \mathcal{O}(y^{1-i_s}) = \mathcal{O}(\kappa y^{1-j})$ , so  $\partial_y^j G = \mathcal{O}(\kappa y^{1-j}) \subseteq \mathcal{O}(y^{1-j})$  for  $j \geq 2$  (and  $\partial_y G = \mathcal{O}(1)$ ). The same Faà di Bruno expansion with  $m \geq 1$  gives the second claim. By Leibniz,  $\partial_y^j [\psi(G)^{N+1}] = \mathcal{O}(\kappa^{N+1} y^{N+1-j})$ , and  $\partial_y^{N-j} [\partial_y G] = \partial_y^{N-j+1} G = \mathcal{O}(y^{-(N-j)})$ . Applying (9.11) with  $\Psi = \text{id}$  and  $n = N+1$ ,

$$\partial_\lambda^{N+1} G = \partial_y^N [\psi(G)^{N+1} \partial_y G] = \sum_{j=0}^N \binom{N}{j} \partial_y^j [\psi(G)^{N+1}] \partial_y^{N-j+1} G = \mathcal{O}(\kappa^{N+1} y)$$

uniformly in  $\lambda$ , and the integral remainder is  $\mathcal{O}(y \kappa(y)^{N+1}) = \mathcal{O}(y^{1+(N+1)\delta} M(y)^{(N+1)M})$ .  $\square$

*Remark 9.6.* Applied to  $\varphi(x) = x^2(1 + \log x)$  ( $\delta = 1$ ,  $M = 1$ ) the theorem re-proves the remainder  $\mathcal{O}(y^{N+2} M^{N+1})$  of (8.5) with a proof that uses no analyticity at all; applied to  $\varphi(x) = x^2 \sin(\log x)$  ( $\delta = 1$ ,  $M = 0$ ) it justifies  $g(y) = y - y^2 \sin L + y^3 \sin L(2 \sin L + \cos L) + \mathcal{O}(y^4)$ ,  $L = \log y$ , a perturbation whose coefficients oscillate and which no power-log expansion describes. The package's `PerturbativeInverse` produces the terms; the theorem is the user's tool for asserting the remainder.

## 10 Finite endpoints, infinity, and powers of the inverse

The branch theorem, coefficient formula and parametric convergence proof only need  $p \neq 0$  and  $u \rightarrow 0^+$ : the substitution  $t = u^p$  and the marker theorem are applied at positive points regardless of the endpoint approached by  $t$ . Statements that use positivity explicitly, such as the exponent-cutoff corollary to the Rouché majorant, retain their stated restrictions. This section makes the endpoint bookkeeping explicit.

### 10.1 Local variables

Let  $f$  be given near  $x_0$ , and let  $u \rightarrow 0^+$  be the local variable of  $x$ :

$$u = \begin{cases} x - x_0 & x \rightarrow x_0^+ \text{ (Direction } \rightarrow \text{ "FromAbove")}, \\ x_0 - x & x \rightarrow x_0^-, \\ 1/x & x \rightarrow +\infty, \\ -1/x & x \rightarrow -\infty. \end{cases} \quad (10.1)$$

Substituting into  $f$  and expanding (Section 3) gives a germ  $y_0 + au^p(1 + \sum u^{\delta_i} B_i(\log u))$  in the sense of (4.1), where  $y_0 = \lim f$  is finite when  $p > 0$ . For  $p < 0$  the target endpoint is  $\pm\infty$  with the sign of  $a$ , while the additive offset  $y_0$  in (4.1) is set to 0; any constant term is a correction with gap  $-p$ . The local variable of  $y$  is

$$w = \begin{cases} y - y_0 & a > 0, p > 0, \\ -(y - y_0) & a < 0, p > 0, \\ 1/|y| & p < 0. \end{cases} \quad (10.2)$$

The uniformizer  $z = (v/a)^{1/p}$  tends to  $0^+$  in all cases. Its relation to  $w$  is  $w = |a|z^p$  for  $p > 0$  and  $w = z^{|p|}/|a|$  for  $p < 0$ .

## 10.2 The theorem at an arbitrary endpoint

**Theorem 10.1** (Signed leading power). *Theorems 4.1, 6.1, 7.1 and 8.2 hold verbatim for the germ (4.1) with  $p < 0$ : the unique branch with  $u \rightarrow 0^+$  is  $u = g(y)$  with  $g \sim z$ ,  $g^r$  has the expansion (6.1) convergent for small  $z$  (i.e. for  $|y|$  large), and the remainder after the exponent cutoff  $q$  in  $w = 1/|y|$  is  $\mathcal{O}(w^{(r+\sigma_*)/|p|} M(w)^{k_*})$ .*

*Proof.* The derivative  $f'(u) = apu^{p-1}(1 + o(1))$  has a constant sign, so  $f$  is monotone on  $(0, \varepsilon)$  with  $f \rightarrow \pm\infty$ ; the inverse exists on a neighbourhood of  $\pm\infty$  and  $u \sim z$ . The proof of Theorem 6.1 is a computation at a fixed positive  $q = v/a$  and never uses the sign of  $p$ . The parametric equation (7.2) has  $\partial_w \mathcal{E}(1, 0) = p \neq 0$  regardless of sign, and the specialisation  $T_{ij} = z^{\delta_i} L^j \rightarrow 0$  as  $z \rightarrow 0^+$  is the same. The remainder theorem is Lemma 3.2.  $\square$

In the variable  $y$  the blocks of  $g^r$  read  $(v/a)^{(r+\sigma)/p} C_\sigma^{(r)} (\frac{1}{p} \log(v/a))$ , with negative exponents of  $y$  when  $p < 0$ ; the exponent of  $w$  is  $(r + \sigma)/|p|$ , which is the quantity the cutoff refers to.

## 10.3 Powers of the inverse and the original variable

The observable power  $r$  in (6.1) gives directly the expansion of  $u^r$ . In terms of  $x$ :

$$\begin{aligned} (x - x_0)^r &= g(y)^r \quad (x \rightarrow x_0^+), & (x_0 - x)^r &= g(y)^r \quad (x \rightarrow x_0^-), \\ x^r &= g(y)^{-r} \quad (x \rightarrow +\infty), & x^r &= (-1)^r g(y)^{-r} \quad (x \rightarrow -\infty), \end{aligned} \quad (10.3)$$

The last equality in (10.3), involving  $(-1)^r$ , is used here only for integer  $r$ . Thus  $x^r$  at infinity is obtained from (6.1) with local-coordinate power  $-r$ . Both signed distances  $x - x_0$  from above and  $x_0 - x$  from below are positive and admit arbitrary real powers. For  $r = 1$  and a finite endpoint, the original inverse is  $x = x_0 \pm g(y)$ . A non-integer power of a negative original variable needs an additional branch convention and is outside the positive-real-power convention of this article.

## 10.4 Examples

**Example 10.2** (The Wright omega function).  $f(x) = x + \log x$  as  $x \rightarrow +\infty$ :  $u = 1/x$ ,  $f = u^{-1} - \log u = u^{-1}(1 - u \log u)$ , so  $a = 1$ ,  $p = -1$ , one gap  $\delta = 1$  with  $B = -L$ , and target endpoint  $+\infty$ . The uniformizer is  $z = 1/y$ ,  $L = -\log y$ , and (6.1) with  $r = -1$  gives the original variable  $x = 1/g(y)$ :

$$x = y - \log y + \frac{\log y}{y} + \frac{\frac{1}{2} \log^2 y - \log y}{y^2} + \frac{\frac{1}{3} \log^3 y - \frac{3}{2} \log^2 y + \log y}{y^3} + \mathcal{O}\left(\frac{\log^4 y}{y^4}\right), \quad (10.4)$$

the classical expansion of the Wright omega function  $\omega(y) = W(e^y)$  [10, 11]; the blocks are the Lambert polynomials  $P_n(\log y)$  of [11].

**Example 10.3** (Reciprocal corrections).  $f(x) = x + 1/x$  at  $+\infty$ :  $u = 1/x$ ,  $f = u^{-1}(1 + u^2)$ ,  $p = -1$ , gap 2, constant coefficient. The inverse is the larger root of  $x^2 - yx + 1 = 0$ ,  $x = (y + \sqrt{y^2 - 4})/2 = y - y^{-1} - y^{-3} - 2y^{-5} - \dots$  (the coefficients are Catalan numbers), which (6.8) reproduces.

**Example 10.4** (Half-integer gaps).  $f(x) = x + \sqrt{x}$  at  $+\infty$ :  $f = u^{-1}(1 + u^{1/2})$ , gap  $\frac{1}{2}$ . The inverse is  $x = (\sqrt{4y + 1} - 1)^2/4 = y - \sqrt{y} + \frac{1}{2} - \frac{1}{8\sqrt{y}} + \frac{1}{128y^{3/2}} + \mathcal{O}(y^{-5/2})$ ; the constant term  $\frac{1}{2}$  is the block of weight  $2 \cdot \frac{1}{2}$ , i.e. exponent 0 in  $w = 1/y$ , despite the infinite target endpoint.

**Example 10.5** (A finite endpoint from below).  $f(x) = x - x^2$  near  $x_0 = 0$  with  $x \rightarrow 0^-$ :  $u = -x$ ,  $f = -u - u^2$ ,  $a = -1$ ,  $p = 1$ ,  $y_0 = 0$ , and  $w = -y$  (the image is  $y < 0$ ). The inverse is  $u = -y - y^2 - 2y^3 - \dots$ , i.e.  $x = y + y^2 + 2y^3 + \mathcal{O}(y^4)$ , the branch of  $x = (1 - \sqrt{1 - 4y})/2$  continued to negative  $y$ .

**Example 10.6** (A pole).  $f(x) = 1 + 1/x$  near  $x_0 = 0^+$ :  $f = u^{-1}(1 + u)$ ,  $p = -1$ , target endpoint  $+\infty$ , and  $x = 1/(y - 1) = y^{-1} + y^{-2} + \dots$  with remainder  $\mathcal{O}(y^{-3})$  after two terms.

**Example 10.7** (The second question, inverted). For  $F(x) = (1 + x + x^{\sqrt{2}})^{\sqrt{2}}$  at  $+\infty$  the forward germ is  $u^{-2}(1 + \sqrt{2}u^{\sqrt{2}-1} + \sqrt{2}u^{\sqrt{2}} + \dots)$  from (1.5), with gaps generated by  $\sqrt{2} - 1$  and  $\sqrt{2}$ ,  $p = -2$ , target endpoint  $+\infty$ , and  $z = y^{-1/2}$ . The inverse is

$$\begin{aligned} x = \sqrt{y} - \frac{1}{\sqrt{2}} y^{(2-\sqrt{2})/2} + \left(\frac{3}{4} - \frac{1}{2\sqrt{2}}\right) y^{(3-2\sqrt{2})/2} \\ + \left(1 - \frac{5}{3\sqrt{2}}\right) y^{(4-3\sqrt{2})/2} - \frac{1}{\sqrt{2}} y^{(1-\sqrt{2})/2} + \mathcal{O}(y^{(5-4\sqrt{2})/2}), \end{aligned} \quad (10.5)$$

the terms being ordered by decreasing exponent  $-(r + \sigma)/2$  of  $y$ , with local-coordinate power  $r = -1$ . The forward remainder is transported by (9.3) with that value of  $r$ ; the package raises the forward order until the cap reaches the requested cutoff. An independent derivation removes the outer power exactly:  $1 + x + x^{\sqrt{2}} = y^{1/\sqrt{2}}$ , then applies the finite-germ coefficient formula. Numerically, with seven blocks at  $y = 10^{16}$  the recorded error is  $4 \cdot 10^{-11}$  against a leading term  $10^8$  (Table 1).

## 11 Worked examples

All expansions in this section were produced by the package and checked by the residual composition of Section 9 and numerically; the displayed logarithms are log of the indicated argument.

### 11.1 The two examples of the first question, once more

For  $f_1 = x + x^2(1 + \log x)$  the package call `AsymptoticInverse[x + x^2 (1 + Log[x]), {x, 0}, {y, 6}]` returns (1.3) with the remainder `PowerLogRemainder[y, 6, 5]`; the sixth block is

$$\begin{aligned} P_5(L) &= -42L^5 - \frac{3727}{12}L^4 - \frac{5365}{6}L^3 - 1256L^2 - \frac{5177}{6}L - \frac{2789}{12} \\ &= -\frac{1}{12}(1 + L)(504L^4 + 3223L^3 + 7507L^2 + 7565L + 2789), \end{aligned}$$

and the seventh  $P_6 = \frac{1}{60}(1 + L)(7920L^5 + 63852L^4 + 200833L^3 + 308537L^2 + 231883L + 68307)$ . With rational exponents the same object is available as `SeriesData[y, 0, {1, -1 - Log[y], 3 + 5 Log[y] + 2 Log[y]^2, ...}, 1, 6, 1]`, on which the built-in series arithmetic works (with the caveat that its `0[y]^6` hides the factor  $M(y)^5$ ).

For  $f_2 = x + x^{\sqrt{2}}$ , `SeriesTermGoal -> 4` returns exactly the four terms requested in the question with remainder `PowerLogRemainder[y, 4 Sqrt[2] - 3, 0]`; the exponent cutoff that produces the same result is  $4\sqrt{2} - 3$ .

### 11.2 Nonunit leading power

$f(x) = 3x^2(1 + x(1 + \log x))$ :  $a = 3$ ,  $p = 2$ ,  $\delta = 1$ ,  $B = 1 + L$ . With  $z = \sqrt{y/3}$  and  $L = \log z = \frac{1}{2} \log(y/3)$ , (6.3) gives

$$g(y) = z - \frac{z^2}{2}(1 + L) + \frac{z^3}{8}(5L^2 + 12L + 7) - \frac{z^4}{48}(\mathcal{D} + 6)(\mathcal{D} + 8)(1 + L)^3 + \mathcal{O}(z^5 M^4), \quad (11.1)$$



where  $\frac{1}{48}(\mathcal{D}+6)(\mathcal{D}+8)(1+L)^3 = (1+L)^3 + \frac{7}{8}(1+L)^2 + \frac{1}{8}(1+L)$ . Two ingredients are easy to forget: the factor  $p^{-n} = 2^{-n}$  and the logarithm  $L = \frac{1}{2} \log(y/3)$ , not  $\log y$ . In the variable  $y$  the expansion reads

$$g(y) = \frac{\sqrt{y}}{\sqrt{3}} - \frac{y}{3} \left( \frac{1}{2} + \frac{\log(y/3)}{4} \right) + \frac{y^{3/2}}{3\sqrt{3}} \left( \frac{7}{8} + \frac{3}{4} \log \frac{y}{3} + \frac{5}{32} \log^2 \frac{y}{3} \right) - \frac{y^2}{9} \left( 2 + \frac{39}{16} \log \frac{y}{3} + \frac{31}{32} \log^2 \frac{y}{3} + \frac{1}{8} \log^3 \frac{y}{3} \right) + \mathcal{O}(y^{5/2} M^4).$$

### 11.3 Several gaps

**Irrational and logarithmic gaps together.**  $f(x) = x + x^{\sqrt{2}} + x^2(1 + \log x)$  has gaps  $\delta_1 = \sqrt{2} - 1$  and  $\delta_2 = 1$ . Below the exponent 3 the support is  $\{1, \sqrt{2}, 2\sqrt{2} - 1, 2, 3\sqrt{2} - 2, 1 + \sqrt{2}, 4\sqrt{2} - 3, 2\sqrt{2}\}$  and

$$g(y) = y - y^{\sqrt{2}} + \sqrt{2} y^{2\sqrt{2}-1} - y^2(1+L) - \frac{6-\sqrt{2}}{2} y^{3\sqrt{2}-2} + y^{1+\sqrt{2}}((\sqrt{2}+2)L + \sqrt{2}+3) + \left( \frac{17\sqrt{2}}{3} - 4 \right) y^{4\sqrt{2}-3} - y^{2\sqrt{2}} \left( (3\sqrt{2}+5)L + 5\sqrt{2} + \frac{13}{2} \right) + \mathcal{O}(y^3 M^2), \quad L = \log y,$$

the mixed blocks coming from (6.4): the index  $(1, 1)$  gives  $(\mathcal{D}+2+\sqrt{2})(1+L) = (\sqrt{2}+2)(1+L)+1$ . The remainder is the block  $y^3(2L^2 + 5L + 3)$  of the index  $(0, 2)$ , of degree 2.

**Two irrational gaps.**  $f(x) = x + x^{\sqrt{2}}(1 + \log x) + 2x^{\sqrt{3}}$  has gaps  $\sqrt{2} - 1$  and  $\sqrt{3} - 1$ , whose sums are the numbers  $k_1(\sqrt{2} - 1) + k_2(\sqrt{3} - 1)$ ; the package represents them by `RootReduce` (for instance  $\sqrt{2} + \sqrt{3} = \text{Root}[1 - 10 \#^2 + \#^4 \&, 4]$ ) and displays them with `ToRadicals`. Below the exponent 3 there are eleven blocks; the residual check passes only when the coefficients, which involve  $\sqrt{5 + 2\sqrt{6}} = \sqrt{2} + \sqrt{3}$ , are canonicalised with `RootReduce`: this is precisely the zero-recognition defect that made one test of report 01 fail (Section 14).

### 11.4 Automatically expanded forward functions

$\sin x + x^2 \log x$  is not a finite power-log sum. The evaluator of Section 3.2 expands it to  $x + x^2 \log x - \frac{1}{6}x^3 + \frac{1}{120}x^5 + \mathcal{O}(x^7)$ , records  $(\rho, k) = (7, 0)$ , and the inverse below the exponent 4 is

$$g(y) = y - y^2 \log y + y^3 \left( \frac{1}{6} + \log y + 2 \log^2 y \right) + \mathcal{O}(y^4 M^3);$$

here the  $y^3$  block combines  $(\mathcal{D}+4)L^2/2 = 2L^2 + L$  from the index  $(2, 0)$  (gap 1,  $B_1 = L$ ) with  $+\frac{1}{6}$  from the index  $(0, 1)$  (gap 2,  $B_2 = -\frac{1}{6}$ ). Similarly  $\tan x$  gives `arctan: SeriesTermGoal -> 4` yields  $y - \frac{1}{3}y^3 + \frac{1}{5}y^5 - \frac{1}{7}y^7$ , the forward order being raised automatically until the transported remainder allows the fourth term.

### 11.5 Symbolic data

With symbolic coefficients the exponents remain numeric and everything goes through under assumptions that make the leading coefficient's sign and the reality of the coefficients provable:  $f = ax + bx^2$  with  $a > 0$ ,  $b \in \mathbb{R}$  gives  $g = y/a - by^2/a^3 + 2b^2y^3/a^5 + \mathcal{O}(y^4)$ . With a symbolic exponent the ordering of the support is undecidable in general and depth truncation (Theorem 8.6) is used:  $f = x + x^\alpha$  with  $\alpha > 1$  gives, at depth 3,

$$g = y - y^\alpha + \alpha y^{2\alpha-1} - \frac{\alpha(3\alpha-1)}{2} y^{3\alpha-2} + \mathcal{O}(y^{4\alpha-3}),$$

which is (6.9) with a symbolic  $\alpha$ .

## 11.6 Powers and logarithms of the inverse

For  $f = x + x^2$ , (6.1) with  $r = 2$  gives  $g^2 = y^2 - 2y^3 + 5y^4 - 14y^5 + \dots$  (Catalan numbers again), with  $r = -1$  it gives  $1/g = y^{-1} + 1 - y + 2y^2 - 5y^3 + \dots$ , and (6.2) gives  $\log g = \log y - y + \frac{3}{2}y^2 - \frac{10}{3}y^3 + \dots$ . These expansions are exact consequences of the same coefficient formula and cost nothing extra; expanding  $\text{Normal}[g]^2$  instead would silently lose the remainder information.

## 12 Lambert coordinates and boundaries of the theory

The convergent power-log class is closed under the operations of Theorem 3.3 and inversion of germs with a monomial leading block. A leading logarithm or exponential changes the natural coordinate. For exact Lambert cores, that coordinate change reduces the problem to the same polynomial-logarithmic arithmetic. More general perturbations require additional bookkeeping, described below.

### 12.1 A leading logarithm: Lambert $W$ and the log log scale

Let  $a \neq 0$ ,  $p \neq 0$ ,  $b \neq 0$ , and  $x \rightarrow 0^+$  with  $x < 1$ . The exact core  $ax^p(-\log x)^b = y$  has  $Y = y/a > 0$ . With  $\tau = -\log x \rightarrow +\infty$ , it reads  $\tau^b e^{-p\tau} = Y$ , hence

$$x = \exp\left[\frac{b}{p} W_k\left(-\frac{p}{b} Y^{1/b}\right)\right], \quad k = \begin{cases} -1, & p/b > 0, \\ 0, & p/b < 0. \end{cases} \quad (12.1)$$

The branch is forced by  $\tau \rightarrow \infty$ : when  $p/b > 0$  the argument approaches 0 from below and  $W_{-1} \rightarrow -\infty$ ; when  $p/b < 0$  it tends to  $+\infty$  and  $W_0 \rightarrow +\infty$ . For  $p, b > 0$ , the other real root on  $W_0$  approaches  $x = 1$  as  $Y \rightarrow 0$ , while the two branches meet at the maximum  $x = e^{-b/p}$ . For instance the inverse of  $x \log(1/x)$  near  $0^+$  is  $-y/W_{-1}(-y)$ , and that of  $x \log x$  on the same source branch is  $y/W_{-1}(y)$  for  $y \rightarrow 0^-$ . For  $p > 0$ , with  $\Lambda = \log(a/y) \rightarrow \infty$  and  $\Lambda_2 = \log \Lambda$ , the expansion begins

$$x = \left(\frac{y}{a}\right)^{1/p} \left(\frac{\Lambda}{p}\right)^{-b/p} \left[1 - \frac{b^2}{p\Lambda} \log \frac{\Lambda}{p} + \mathcal{O}\left(\frac{\Lambda_2^2}{\Lambda^2}\right)\right], \quad (12.2)$$

a series in  $1/\Lambda$  with polynomial coefficients in  $\Lambda_2$ , multiplying the leading factor. The following construction computes all its logarithmic orders, with convergence and a remainder bound.

### 12.2 A convergent universal Lambert expansion

Let  $Z$  approach  $+\infty$  on the principal real branch or  $0^-$  on the lower real branch. Set

$$s = \begin{cases} 1, & W_0(Z), \ Z \rightarrow +\infty, \\ -1, & W_{-1}(Z), \ Z \rightarrow 0^-, \end{cases} \quad A = |\log |Z||, \quad t = A^{-1}, \quad \ell = \log t.$$

Then  $\log |W| + W = \log |Z| = sA$ . Writing  $W = s(1 + V)/t$  gives the normalized equation

$$V + st \log(1 + V) - st\ell = 0. \quad (12.3)$$

**Theorem 12.1** (Lambert expansion in a logarithmic coordinate). *Equation (12.3) has a unique solution  $V \rightarrow 0$  of the form  $\sum_{n \geq 1} t^n P_n(\ell)$  with  $P_1 = s\ell$  and  $\deg P_n \leq n-1$  for  $n \geq 2$ . This series converges for sufficiently small  $t > 0$ . For  $N \geq 1$ , truncating  $V$  after  $t^N$  leaves  $\mathcal{O}(t^{N+1}M(t)^N)$ , and multiplying by  $s/t$  gives the corresponding remainder for  $W$ . In particular,*

$$W = \frac{s}{t} + \ell - s\ell t + \left(\ell + \frac{\ell^2}{2}\right)t^2 - s\left(\ell + \frac{3\ell^2}{2} + \frac{\ell^3}{3}\right)t^3 + \mathcal{O}(t^4 M(t)^4). \quad (12.4)$$

*Proof.* Replace  $t\ell$  by an independent parameter  $T$ . The function  $V + st\log(1 + V) - sT$  is analytic near  $(V, t, T) = (0, 0, 0)$  and its  $V$ -derivative there is 1. The analytic implicit function theorem gives a convergent Taylor series  $V(t, T)$ . Moreover  $V = sT + tH(t, T)$  for an analytic  $H$  with  $H(0, 0) = 0$ . After  $T = t\log t$ , every term of total degree  $n \geq 2$  therefore has logarithmic degree at most  $n - 1$ , and  $P_1 = s\ell$ . Cauchy estimates bound the coefficient norms exponentially. The tail starting at degree  $N + 1$  is bounded by a geometric series with first term  $t^{N+1}M(t)^N$ , proving the remainder. Coefficients can be computed by the iteration  $V \leftarrow st\ell - st\log(1 + V)$ , which gains one power of  $t$  each time, or by Newton iteration. The real solution has  $W \sim sA$  and hence belongs to the specified real Lambert branch. Expanding the equation gives (12.4).  $\square$

An explicit sufficient convergence condition is

$$q_L = 4t(1 + |\log t|) < 1. \quad (12.5)$$

Indeed, for complex  $|t|, |T| \leq 1/4$ , the map  $V \mapsto sT - st\log(1 + V)$  sends  $|V| \leq 1/2$  into itself, since  $(1 + \log 2)/4 < 1/2$ , and has derivative bounded by  $1/2$ . Its fixed point is jointly analytic and bounded by  $1/2$ , so its Taylor coefficients are bounded by  $4^{a+b}/2$ . Beyond the leading term  $sT$ , every coefficient has  $a \geq 1$  because  $V(0, T) = sT$ . If  $V_N$  retains total degrees through  $N \geq 1$ , the geometric majorant therefore gives

$$|V - V_N| \leq \frac{2t q_L^N}{1 - q_L}, \quad \left| W - \frac{s}{t}(1 + V_N) \right| \leq \frac{2q_L^N}{1 - q_L}. \quad (12.6)$$

The construction follows the two-small-parameter method in the supplied transseries notes [11]; it establishes a sufficient domain and does not claim convergence from a fixed-logarithm implicit-function argument alone.

For the leading-log core, put  $Z = -(p/b)Y^{1/b}$  and choose  $s$  from its branch. Since  $-bs/p > 0$ , an arbitrary real observable power is best assembled as

$$x^r = Y^{r/p} \left( -\frac{bs}{p} \right)^{-br/p} t^{br/p} (1 + V)^{-br/p}. \quad (12.7)$$

This uses a positive real power of the unit  $1 + V$ . Directly exponentiating a truncated  $W$  would shift the required precision by one order; (12.7) makes that precision shift explicit.

**Example 12.2** (The inverse of  $x \log x$ ). For  $y \rightarrow 0^-$  let  $A = \log(1/(-y))$  and  $B = \log A$ . The branch with  $x \rightarrow 0^+$  is

$$x = \frac{-y}{A} \left[ 1 - \frac{B}{A} + \frac{B^2 - B}{A^2} + \frac{-B^3 + \frac{5}{2}B^2 - B}{A^3} + \mathcal{O}\left(\frac{(1+B)^4}{A^4}\right) \right].$$

The endpoint  $x \rightarrow 1$  selects a different inverse germ and is not represented by this expansion.

### 12.3 Exponential cores at infinity

For a positive source coordinate  $x$  with  $X = x^p \rightarrow +\infty$ , consider  $y = ax^b e^{cx^p}$ , where  $a \neq 0$ ,  $p \neq 0$ ,  $c \neq 0$ , and  $Y = y/a > 0$ . If  $b \neq 0$ , taking the power  $p/b$  gives

$$X = \frac{b}{cp} W_k \left( \frac{cp}{b} Y^{p/b} \right), \quad k = \begin{cases} 0, & cp/b > 0, \\ -1, & cp/b < 0. \end{cases} \quad (12.8)$$

These branch choices ensure  $X > 0$  and  $X \rightarrow \infty$ : the argument tends to  $+\infty$  in the first case and  $0^-$  in the second. If  $b = 0$ , inversion is elementary,  $x = (\log Y/c)^{1/p}$ . In the Lambert cases, with  $s, t, V$  as above,

$$x^r = \left( \frac{bs}{cp} \right)^{r/p} t^{-r/p} (1 + V)^{r/p},$$

whose positive prefactor and unit series again have explicit precision tracking.

**Example 12.3** (The inverse of  $xe^x$  at infinity). Here  $x = W_0(y)$ . With  $A = \log y$  and  $B = \log A$ ,

$$x = A - B + \frac{B}{A} + \frac{B(B-2)}{2A^2} + \frac{B(2B^2-9B+6)}{6A^3} + \mathcal{O}\left(\frac{(1+B)^4}{A^4}\right).$$

Both this expansion and the  $x \log x$  expansion use  $t = 1/A$  as the local variable. An exponent cutoff in  $t$  specifies logarithmic accuracy; it is not a cutoff in  $y$  or  $1/y$ . A nonconstant external factor such as  $-y$  must also multiply the remainder descriptor.

## 12.4 Perturbations of a Lambert core

Relative perturbations  $F(x) = ax^p(-\log x)^b(1 + \sum_i x^{\delta_i} B_i(\log x))$  can be approached with the general-core formula (5.5). If  $\phi$  is the exact core inverse and  $\tau = -\log \phi(y)$ , then

$$\phi'(y) = \left[ ax^{p-1} \tau^{b-1} (p\tau - b) \right]_{x=\phi(y)}^{-1}.$$

The marker coefficients have powers of  $\phi(y)$  and rational functions of  $\tau$ , with denominators arising from  $p\tau - b$ . A convergent marker expansion requires analytic control on the chosen branch, and re-expanding its coefficients introduces a second order of truncation: perturbation power and logarithmic depth. The exact-core Lambert construction proves the logarithmic expansion without asserting support for arbitrary perturbed cores.

The leading logarithmic polynomial itself can also be generalized before introducing a second perturbation scale.

**Corollary 12.4** (An arbitrary polynomial at the leading exponent). *Let  $F_0(u) = au^p Q(\log u)$ , where  $p \neq 0$  and  $Q$  is a nonconstant real polynomial of degree  $q \geq 1$ , on the real branch  $u \rightarrow 0^+$ . Its inverse has a convergent relative power-log expansion in a reciprocal-logarithm coordinate, even when  $Q$  is not a power of an affine polynomial.*

*Proof.* Absorb the leading coefficient of  $Q$  into  $a$ , so  $Q$  is monic, and set  $b = [L^{q-1}]Q/q$  and  $\tau = -\log u - b$ . Then

$$Q(-\tau - b) = (-\tau)^q R(1/\tau), \quad R(v) = 1 + \sum_{j=2}^q r_j v^j.$$

With  $Y = y/(a(-1)^q) > 0$  (after subtracting any constant target offset),  $k = -p/q$ ,  $s = \text{sign } k$  and  $Z = kY^{1/q}e^{pb/q}$ , take  $A = s \log |Z|$ ,  $t = 1/A$ , and  $\tau = (1+V)/(|k|t)$ . The inverse equation becomes

$$V + st \left[ -\ell + \log(1+V) + \frac{1}{q} \log R\left(\frac{|k|t}{1+V}\right) \right] = 0. \quad (12.9)$$

Replacing  $t\ell$  by an independent parameter again gives an analytic equation with  $V$ -derivative 1 at the origin. Its solution is jointly analytic, and  $R(0) = 1$  makes its real powers analytic too. For any real local-coordinate observable power  $r$ ,

$$u^r = Y^{r/p} \left( \frac{A}{|k|} \right)^{-qr/p} (1+V)^{-qr/p} R\left(\frac{|k|t}{1+V}\right)^{-r/p}.$$

This yields convergence and the same type of finite-order logarithmic remainder as before. The sufficient numerical bound (12.6) was proved for  $R = 1$  and is not asserted unchanged for a general  $R$ .  $\square$

The polynomial correction modifies the logarithmic coefficients and must be included in the equation; it is not beyond all logarithmic orders. In general this inverse has no claimed exact `ProductLog` expression. The construction also applies to logarithmic cores at either infinite source endpoint after  $u = 1/x$  or  $u = -1/x$ , using the observable and sign rules of Section 10.

**Proposition 12.5** (Perturbations beyond every logarithmic order). *Suppose a finite leading-log core  $F_0(u) = au^p Q(\log u)$  from Corollary 12.4, or an affine-log power core, is supplemented by finitely many polynomial-logarithmic terms of strictly larger  $u$ -exponent. Alternatively, suppose a growing exponential core  $F_0(x) = ax^b e^{cx^p}$  with  $c, p > 0$  at  $x \rightarrow +\infty$  is supplemented by a finite power-log sum. Let  $g_0$  and  $g$  be the corresponding inverse branches, with any common target offset subtracted, and let  $t$  be the Lambert coordinate of  $g_0$ . For every fixed real  $r$  and every  $N > 0$ ,*

$$g(y)^r - g_0(y)^r = o(g_0(y)^r t^N).$$

*Consequently every finite relative logarithmic expansion of the core inverse is also an expansion of the full inverse, with the same finite-order remainder class.*

*Proof.* In the leading-log case  $F'_0(u) \sim pF_0(u)/u$ , and the perturbation and its derivative are smaller by a factor  $\mathcal{O}(u^\delta M(u)^D)$  for some  $\delta > 0$  and finite  $D$ . A mean-value bracket near  $g_0$  yields  $(g - g_0)/g_0 = \mathcal{O}(g_0^\delta M(g_0)^D)$ . Here  $1/t$  is a positive constant times  $|\log g_0|$  plus lower logarithmic terms, so this bound is  $o(t^N)$  for every  $N$ .

In the exponential case  $F'_0/F_0 \sim cpx^{p-1}$ , whereas the additive perturbation  $R$  and its derivative have only power-log growth. A bracket of width  $\mathcal{O}(|R(g_0)|/|F'_0(g_0)|)$  has slope comparable to  $F'_0(g_0)$ : the logarithm of that derivative changes by  $\mathcal{O}(R/F_0) = o(1)$  across the bracket. Thus  $(g - g_0)/g_0$  is exponentially small in  $g_0^p$ , while  $t$  is comparable to  $g_0^{-p}$ . Finally apply the mean value theorem to the positive real power  $u^r$ . Continuous local inverse branches exist because the perturbation derivative is smaller than the nonzero core derivative.  $\square$

The package marks this situation with `"LeadingCoreOnly" -> True`, stores the discarded expression in `"BeyondLogarithmicOrders"`, and leaves `"ExactInverseExpression"` unavailable. Those terms affect an exponentially smaller scale even though they do not change any finite coefficient in the reported logarithmic expansion.

## 12.5 Zero gaps: corrections of relative size $1/\log$

For  $f(x) = x + x/\log x$  the correction has power gap 0 and relative size  $1/\log x$ . All corrections of the inverse then sit at the same power  $y$  with increasing negative powers of  $\log y$ , the index set  $I_H$  is infinite, and the exponent cutoff has no meaning. The right coordinate is  $\tau = 1/\log y$ : with  $g = yW$  the equation becomes  $W(1 + \tau/(1 + \tau \log W)) = 1$ , analytic near  $(W, \tau) = (1, 0)$ , so  $W$  is an ordinary convergent power series in  $\tau$ ,  $W = 1 - \tau + \tau^2 - 2\tau^3 + \frac{7}{2}\tau^4 - \dots$ , i.e.  $g(y) = y(1 - 1/\log y + 1/\log^2 y - 2/\log^3 y + \dots)$ . The general-core formula produces these terms, but the scale is the reciprocal-logarithm scale, not the power-log scale, and a package extension needs  $\tau$  as a second generator with its own valuation.

## 12.6 Flat terms

$e^{-1/x}$  is smaller than every power of  $x$ ; a germ  $x + e^{-1/x}$  has the same asymptotic power-log expansion as  $x$ . Its inverse obeys  $g = y - e^{-1/g}$ , so  $y - e^{-1/y} \leq g < y$  and  $(y - g)/y^2 \rightarrow 0$ . Consequently  $e^{-1/g}/e^{-1/y} \rightarrow 1$ , proving  $g - y = -e^{-1/y}(1 + o(1))$ . The marker formula begins  $y - e^{-1/y} + y^{-2}e^{-2/y} + (y^{-3} - \frac{3}{2}y^{-4})e^{-3/y} + \dots$ . These exponential sectors are invisible to a pure power-log expansion. They require a different extension from the exact exponential cores above, whose inverses reduce to Lambert coordinates.

## 12.7 Oscillatory coefficients

$x^2 \sin(\log x)$  is  $\mathcal{O}(x^2)$  with all derivatives of the right size, so Theorem 9.5 applies and the marker formula gives a valid expansion  $y - y^2 \sin L + y^3 \sin L(2 \sin L + \cos L) + \mathcal{O}(y^4)$  whose “coefficients” are bounded oscillating functions of  $L = \log y$ . They are not polynomials in  $L$ , the coefficient algebra is different, and the sharp remainder theory of Section 8 (which uses degrees) does not apply; only the tame theorem does. The package’s `PerturbativeInverse` generates the terms and leaves the remainder to the user.

## 12.8 Symbolic exponents and nonuniformity

When a gap  $\delta$  is a symbol, the order of the weights  $k \cdot \delta$  depends on the parameter: the order of  $2\delta_1$  and  $\delta_2$  changes at  $\delta_2 = 2\delta_1$ . Exponent truncation needs provable comparisons; depth truncation (Theorem 8.6) only needs positive gaps after a valid leading term and real branch have been selected. Nothing is uniform as a gap tends to 0: the  $y$ -radius  $R_\alpha^{1/(\alpha-1)}$  of the  $x + x^\alpha$  inverse tends to 0 as  $\alpha \downarrow 1$ , although  $R_\alpha \rightarrow 1$ , and at  $\alpha = 1$  the inverse is  $y/2$ .

## 12.9 Nested powers of the logarithm

Non-integer powers  $(-\log x)^s$ ,  $\log(-\log x)$ , and negative powers of  $\log x$  in a correction leave the polynomial coefficient algebra. For example  $\varphi(x) = x^{1+\delta}(-\log x)^{1/2}$  is real for small positive  $x$  and satisfies Theorem 9.5 with  $M = 1$ , but repeated differentiation produces negative half-integer as well as positive half-integer powers of  $-L$ . A suitable coefficient algebra therefore contains  $(-L)^{1/2}$  and its inverse. The Euler identity and composition recurrence still apply when the coefficient functions are differentiable or analytic as needed; closure and remainder bounds in the enlarged scale must also be established.

## 13 The package

The package `AsymptoticInverse` (directory `AsymptoticInverse/` of the repository; usage in its `README.md`) implements the power-log engines of Sections 2–10, the Lambert-coordinate construction of Section 12.2, and the logarithmic target transformation of Section 15. Its interface is

|                                                         |                                                    |
|---------------------------------------------------------|----------------------------------------------------|
| <code>AsymptoticInverse[f, {x, x0}, {y, cutoff}]</code> | <code>(* or ..., y, SeriesTermGoal -&gt; n</code>  |
| <code>*)</code>                                         |                                                    |
| <code>AsymptoticExpansion[f, {x, x0, cutoff}]</code>    | <code>(* or {x, x0}, SeriesTermGoal -&gt; n</code> |
| <code>*)</code>                                         |                                                    |

with  $x_0$  a real number, `Infinity` or `-Infinity`, options `Assumptions`, `Direction`, `Method` (“Lagrange” or “Newton” for the power-log engine), `Power` ( $r$ ), `Truncation` (“Exponent” or “Depth”), `InputRemainder` and `MaxTerms`. Both return a `PowerLogSeries` object: `Normal` gives the finite expression, `s["Remainder"]` its inert descriptor, `s["FrontierTerm"]` the computed first omitted block, `s["Terms"]` exponent/coefficient pairs, `s["SeriesData"]` a `SeriesData` object when the scale permits it, and `s["Properties"]` the full property list.

The auxiliary functions provide residual checking through `InverseResidual`, numerical comparison through `InverseNumericalCheck`, marker formulas through `PerturbativeInverse`, individual coefficients through `InverseExpansionCoefficient`, and normalized model data through `PowerLogModel`.

Recognized Lambert cores dispatch to `"Method" -> "Lambert"`. Their objects carry



"Scale" -> "Logarithmic", an exact "Prefactor", and "LogarithmicVariable" -> t. A pair  $\{n, C\}$  contributes  $\text{Prefactor } t^n C(\log t)$ , and a cutoff  $q$  retains  $n < q$  in this normalized bracket. The remainder is  $|\text{Prefactor}| \text{PowerLogRemainder}[t, \beta, k]$ ; the prefactor must not be lost when estimating errors. "LambertBranch" and "TargetDomain" record the selected real branch. Such objects have no ordinary SeriesData in  $y$ , and depth truncation in the original power gaps does not apply. For example,

```
AsymptoticInverse[x Log[x], {x, 0}, {y, 4}]
AsymptoticInverse[x Exp[x], {x, Infinity}, {y, 5}]
```

produce the displayed logarithmic orders in Section 12.2. The first uses  $y \rightarrow 0^-$  and the lower Lambert branch; the second uses  $y \rightarrow +\infty$  and the principal branch.

### 13.1 Data structures and exact comparisons

A jet is a sorted list of pairs  $\{\sigma, C_\sigma\}$ , with exact exponents and polynomial coefficients. Algebraic exponents have **RootReduce** canonical forms, so  $\sqrt{2} + \sqrt{3}$  and  $\sqrt{5 + 2\sqrt{6}}$  are recognized as equal. For other exact exponents, symbolic simplification must prove the comparisons used by the computation; it is not a complete equality algorithm for arbitrary transcendental expressions. Undecidable order comparisons produce a **Failure**. Inexact **Real** input is rejected because a rounded exponent does not identify the exact semigroup: choosing a rational decimal approximation to  $\sqrt{2}$  changes both the problem and its possible resonances.

Coefficients are canonicalised too: algebraic numbers with **RootReduce**, logarithms of positive rationals into sums of logarithms of primes, symbolic parameters with **Simplify** under the assumptions. Without this, exactly vanishing residual coefficients such as  $-2\sqrt{2} - 2\sqrt{3} + 2\sqrt{5 + 2\sqrt{6}}$  or  $-\frac{1}{16} \log^2 4 - \frac{1}{64} \log 2 \log 32 + \frac{1}{64} \log 8 \log 128$  are not recognised as zero by **Simplify**, which is what caused the two test failures among the nine reports (Section 14).

### 13.2 The engines

**Forward.** The evaluator of Section 3.2 is a structural recursion returning triples  $(T, P, D)$ . Finite power-log sums are recognized as exact. The working order grows when cancellation hides the leading term or prevents the requested precision from being reached. Analytic heads are expanded in their small argument, so expressions such as  $\sin(x \log x)$  are handled without asking **Series** to order their full composition. Laurent and Puiseux expansions of supported heads are composed in the same way, with the local increment's sign checked at real branch points. Absolute values use the eventual sign of the leading logarithmic polynomial.

**Inverse, Lagrange.** The forward germ is normalised (Section 4); the multi-indices of weight  $< H$  are enumerated by a depth-first search with exact comparisons ((8.1); the one-step boundary is formed at the same time); for each index the polynomial loop

```
q = Expand[Times @@ MapThread[Power, {polys, k}]];
Do[q = Expand[D[q, ell] + (r + w + p j) q], {j, 1, n - 1}];
q = (-1)^n r q / (p^n Times @@ (Factorial /@ k))
```

evaluates (6.1); equal weights are merged and the blocks of weight  $\sigma_*$  on the boundary are summed into the frontier term. For **SeriesTermGoal** -> n, a persistent frontier visits successive complete weights until  $n$  nonzero blocks exist, reusing previous coefficients. The independent "GroupedLagrange" method combines collisions before applying the Euler operators; Section 16 derives both changes.



**Inverse, Newton.** The normalised equation (6.11) and its derivative are evaluated with the composition recurrence (Lemma 2.7), and the derivative’s reciprocal by the geometric series of a unit with constant leading term. The working cutoff doubles through exact weights until it reaches  $H$ ; each recorded precision has a verified formal residual. The three ordinary engines are compared in the test suite.

**Lambert.** The separate `Kernel/LambertInverse.wl` module recognizes affine-log powers, arbitrary leading logarithmic polynomials and exponential–power cores. It solves (12.3) or its polynomial correction (12.9) using unit-series arithmetic, then forms the observable bracket. Logarithmic cores at finite endpoints and either infinity use the same positive source coordinate. For an arbitrary leading polynomial, metadata does not claim an exact `ProductLog` inverse. The original forward expression is retained for numerical checks. The restricted perturbations of Proposition 12.5 carry metadata identifying the leading-core approximation.

**Assembly.** Blocks  $z^{r+\sigma}C(L)$  are converted to  $(v/a)^{(r+\sigma)/p}C(\log(v/a)/p)$ , and the sign and shift of (10.3) are applied. The forward remainder is transported with exponent  $\rho - p + r$  in the local source variable; at infinite source endpoints, the requested original-variable power changes the local observable to  $-r$ . This observable dependence is essential for a valid cap on the requested cutoff.

### 13.3 Checks

`InverseResidual` composes the stored finite forward model with the stored blocks and checks whether  $(f(G) - y_0)/(az^p) - 1$  vanishes below the relative cutoff. A larger cutoff can expose the first omitted block and its factor  $-p$  from (9.6). This is independent of the coefficient formula, but its scope is the finite model. Lambert objects use the normalized Lambert equation and identify that scope separately. `InverseNumericalCheck` solves the original  $f(x) = y_1$  at high precision, checks the selected real source side, and reports the approximation error and remainder scale; numerical agreement does not prove an asymptotic bound. The test runner `Tests/RunTests.wl` combines the original examples with review regressions, expanded-input cases, Lambert cases and performance regressions.

### 13.4 Cost

The retained multi-indices number at most  $\prod_i(1 + \lfloor H/\delta_i \rfloor)$ , and each coefficient uses  $|\mathbf{k}| - 1$  bidiagonal polynomial operations. Sparse multiplication now skips product pairs already beyond the cutoff; nonnegative integer powers use repeated squaring; and depth enumeration visits the simplex  $|\mathbf{k}| \leq N$  and its boundary directly instead of constructing a Cartesian box. Resource limits are checked while indices are generated. Exact algebraic canonicalization and sparse composition remain significant costs, and Newton’s logarithmic iteration count alone does not predict which method is faster.

The reproducible `Tests/BenchmarkPerformance.wl` compares these changes with frozen reference algorithms and checks equality of their results. A Windows/Wolfram run recorded the following seconds; these are fixture measurements, not portable speed guarantees.

| Fixture                         | Reference | Updated |
|---------------------------------|-----------|---------|
| 200 × 200 product, cutoff 5     | 2.90063   | 0.02723 |
| Exact monomial to power 2048    | 2.08708   | 0.01991 |
| Depth 2 in 10 dimensions        | 4.14531   | 0.03753 |
| Weights {1, 2, 3, 5}, cutoff 20 | 0.23768   | 0.41552 |

The last fixture is slower after bounded allocation; the improvements target wasted product work, repeated multiplication and Cartesian depth enumeration rather than guaranteeing a speedup for every input.

## 14 The nine reports

Nine independently written reports on the first question [12] preceded this article; each contains an article of 22–30 pages, a Wolfram Language package, native tests, and a Python verification record, and none of them had been run in a Wolfram kernel by its authors. Their detailed comparison is in `reports/COMPARISON.md`; this section summarises what they agree on, what each adds, and what was wrong.

### 14.1 Agreement

All nine treat the finite germ (4.1) (report 07 with  $p = 1$  only), select the real branch by  $x/(y/a)^{1/p} \rightarrow 1$ , prove the coefficient formula (6.1) with exactly the same normalization through the marker form of Lagrange–Bürmann inversion, prove convergence (four by the parametric implicit function theorem, five by Rouché’s theorem with Cauchy estimates), prove a remainder theorem with the logarithmic factor, derive the transport of forward remainders, and compute the same expansions (1.3)–(1.4) with the Catalan and Fuss–Catalan laws. All represent expansions as sparse exact jets, reject floating-point input, and provide a residual check.

### 14.2 Individual contributions absorbed here

- Report 01: the residue proof of the marker theorem and its formal clause; the  $g^r$  and  $\log g$  formulas; the `InputRemainder` interface with cutoff cap; the  $x + x/\log x$  analysis.
- Report 02: the observable power  $r$  built into the main formula and implemented; the Picard fixed-point engine; the explicit marker-disc convergence criterion; the partial resummation in  $y \log y$ .
- Report 03: the exact degree and leading-coefficient formula; the contraction proof of convergence; the search bound  $\lceil B/\delta_* \rceil \delta_*$  for the frontier.
- Report 04: the first omitted block as an asymptotic equality with its complete polynomial; the successive-power engine; the condition number.
- Report 05: the tame near-identity theorem (here Theorem 9.5, with a complete proof); Picard iteration; global bounds for both examples.
- Report 06: the composition recurrence (Lemma 2.7); the Newton engine in the jet algebra; the rising-factorial closed form (6.8).
- Report 07: the explicit conditional majorant  $\mathcal{Q}_r$  (Theorem 7.3) and its cutoff form; the negative-side germ by reflection.
- Report 08: the explanation of the complex germ (1.2); the Lambert- $W$  core (12.1) and the log log scale; the finite-point slope bound; flat perturbations.
- Report 09: depth truncation with symbolic exponents; the harmonic-number formula (6.7); the explicit tail majorant for the log example.

### 14.3 Errors found

Report 07 asserts that a nonunit leading power forces an infinite binomial expansion; the substitution  $t = x^p$  in Theorem 6.1 shows that the coefficient formula stays finite and exact. Reports 07 and 08 state remainder degrees  $(d[\sigma_*/\delta], Nd)$  that are valid but can be far from the exact degree of the frontier block. Report 05's proof of the tame theorem omits the inductive estimates of Step 4 of Theorem 9.5. Two residual checkers (reports 01 and 04) rely on `Simplify` to recognise zero and fail on denested radicals and on logarithms of composite integers.

### 14.4 Native execution

Running the nine test suites on Wolfram 15.0.1 gave: 01: 35/36; 02: 95/95; 03: 34/34; 04: 34/35; 05: 38/38; 06: 33/33; 07: 34/34; 08: 40/40; 09: 35/35. The two failures are the zero-recognition defects just mentioned, in residual checkers, not wrong expansions. Every package therefore computes what its article claims, which is a strong independent confirmation of the formula (6.1) and of the two answers. What none of them does, and what motivated the present package, is to treat finite endpoints from both sides and infinity through signed leading powers, to expand arbitrary forward functions with a rigorous remainder (which is the second question), to track precision and logarithmic degree through compositions, and to return `SeriesData` when the exponents allow it.

## 15 Inversion through explicit coordinate transformations

The leading expression in the original variable need not belong to the scale used by the inverse engine. A change of source or target coordinate can put the equation into a supported scale while preserving the requested real branch. The transformation is part of the mathematical result: it determines the expansion variable, how the cutoff is interpreted, and how an inner remainder becomes an error in the original variable.

### 15.1 Target transformations

**Theorem 15.1** (An exact change of target coordinate). *Let  $F$  be continuous and strictly monotone on a source interval  $I$ , and let  $J$  be continuous and one-to-one on  $F(I)$ . Put  $\Phi = J \circ F$ . If  $h$  is the inverse of  $\Phi$  on  $J(F(I))$ , then the inverse of  $F$  on  $F(I)$  is*

$$g(y) = h(J(y)).$$

*If an observable of that inverse has an expansion  $h(v)^r = A(v) + \mathcal{O}(R(v))$  as  $v$  approaches the transformed target endpoint, then*

$$g(y)^r = A(J(y)) + \mathcal{O}(R(J(y)))$$

*on the corresponding source branch, provided the same real power is defined there.*

*Proof.*  $\Phi(h(J(y))) = J(y)$  gives  $J(F(h(J(y)))) = J(y)$ . Injectivity of  $J$  gives  $F(h(J(y))) = y$ , and the source interval fixes the inverse branch. The remainder statement is direct substitution along the transformed endpoint; no output reconstruction is involved.  $\square$

The theorem distinguishes two operations that are sometimes conflated. Changing the target of the equation does not amplify the error of its inverse observable. Changing the source variable and then reconstructing it generally does, as discussed below.

## 15.2 Exponential phases

Suppose an expression can be written on the selected real source branch as

$$F(x) = y_0 + \varepsilon e^{\Phi(x)}, \quad \varepsilon \in \{1, -1\}. \quad (15.1)$$

An eventually signed amplitude is included in the phase: if the original expression is  $y_0 + A(x)e^{P(x)}$ , prove the eventual sign  $\varepsilon$  of  $A$ , then set  $\Phi = P + \log(\varepsilon A)$ . The inverse equation becomes

$$\Phi(x) = v, \quad v = \log(\varepsilon(y - y_0)), \quad \varepsilon(y - y_0) > 0. \quad (15.2)$$

The phase may contain several powers and polynomial–logarithmic terms even when the original function has exponential growth or decay. It is inverted on the same source interval. If  $\Phi \rightarrow +\infty$ , then  $F \rightarrow \varepsilon\infty$ ; if  $\Phi \rightarrow -\infty$ , the target approaches  $y_0$  from the side with sign  $\varepsilon$ .

**Proposition 15.2** (The original residual and the phase residual). *For a real approximation  $X$  on that source interval, put  $D = \Phi(X) - v$ . Then*

$$\frac{F(X) - y}{y - y_0} = e^D - 1. \quad (15.3)$$

*In particular, if  $D \rightarrow 0$ , the relative original residual is  $D + \mathcal{O}(D^2)$ . If  $|\Phi| \geq \mu > 0$  between  $X$  and the true inverse  $g(y)$ , then  $|X - g(y)| \leq |D|/\mu$ .*

*Proof.* Since  $y - y_0 = \varepsilon e^v$ , divide  $\varepsilon(e^{\Phi(X)} - e^v)$  by  $\varepsilon e^v$ . The remaining statements follow from the Taylor expansion of  $e^D$  at 0 and the mean value theorem for  $\Phi$ .  $\square$

Thus a phase residual can be numerically useful without forming enormous exponentials. A numerical solve in the phase remains numerical evidence unless the derivative and residual have rigorous bounds; it nevertheless refers to the original equation through the exact identity (15.3).

**Example 15.3** (Several powers in an exponential). For  $F(x) = e^{x^2+x}$  at  $+\infty$ , put  $v = \log y$ . The inverse is

$$x = \frac{-1 + \sqrt{1 + 4v}}{2} = \sqrt{v} - \frac{1}{2} + \frac{1}{8\sqrt{v}} - \frac{1}{128v^{3/2}} + \mathcal{O}(v^{-5/2}).$$

This is an ordinary inverse expansion in the phase variable  $v$ . Its local target variable is  $1/v = 1/\log y$ , and its remainder is measured in that coordinate. For  $e^{-x^2-x}$  at the same source endpoint, replace  $v$  by  $-\log y$  and take  $y \rightarrow 0^+$ .

**Example 15.4** (A nonconstant amplitude). For  $F(x) = xe^{x^2+x}$  at  $+\infty$ , the phase is  $\Phi = x^2 + x + \log x$ . In terms of  $v = \log y$ , inversion begins

$$x = \sqrt{v} - \frac{1}{2} + \frac{\frac{1}{8} - \frac{1}{4}\log v}{\sqrt{v}} + \mathcal{O}(v^{-1}(1 + |\log v|)^2).$$

The logarithm of the amplitude contributes at the displayed order. Omitting it would preserve the leading term but give the wrong next coefficient. Translating  $x$  to  $x - c$  translates the reconstructed inverse by  $c$  and leaves the phase calculation in that shifted source coordinate.

**Example 15.5** (A flat forward function with an elementary inverse).  $F(x) = e^{-1/x}$  on  $x \rightarrow 0^+$  has no nonzero power–log forward expansion, but its inverse is exactly

$$x = -\frac{1}{\log y}, \quad 0 < y < 1.$$

The logarithmic target transformation turns the equation into  $-1/x = \log y$ . This example does not require an expansion into flat perturbation sectors: the exponential is part of the exact coordinate transformation.

Variable powers can enter the same route when their real domain is established. For  $x^x$  on  $x > 0$ , the phase is  $x \log x$ , so its inverse at  $+\infty$  reduces to the Lambert-coordinate inversion of that phase. The identity  $A(x)^{B(x)} = \exp(B(x) \log A(x))$  is used only where  $A(x) > 0$ .

### 15.3 Source reconstruction and its error

**Theorem 15.6** (A change of source coordinate). *Let  $x = H(t)$  be a continuously differentiable one-to-one parametrization of the selected source branch, and write  $K(t) = F(H(t))$ . Let  $h$  be its inverse and suppose*

$$h(y) = A(y) + E(y), \quad E(y) = \mathcal{O}(R(y)).$$

*If  $H$  is defined on the interval between  $A$  and  $A + E$ , and  $|H'| \leq B(y)$  there, then*

$$F^{-1}(y) = H(A(y)) + \mathcal{O}(B(y)R(y)).$$

*The same conclusion applies to an observable after replacing  $H$  by that observable composed with  $H$ .*

*Proof.* The branch identity gives  $F^{-1} = H \circ h$ . Apply the mean value theorem between  $A$  and  $h = A + E$ .  $\square$

For the coordinate transformations used here, more informative exact identities specify when a small error survives reconstruction.

**Corollary 15.7** (Exponential reconstruction). *For a positive source coordinate  $u = e^{st}$ ,  $s \neq 0$ , if  $h = A + E$  and  $E \rightarrow 0$ , then*

$$u = e^{sA}(1 + \mathcal{O}(E)), \quad u^r = e^{rsA}(1 + \mathcal{O}(E)) \quad (15.4)$$

*for every fixed real  $r$ . If  $E$  does not tend to zero, an additive expansion for  $h$  alone does not justify these relative error assertions.*

*Proof.*  $u = e^{sA}e^{sE}$ , and  $e^{sE} = 1 + \mathcal{O}(E)$  when  $E \rightarrow 0$ . Replace  $s$  by  $rs$  for the observable.  $\square$

**Corollary 15.8** (Logarithmic reconstruction). *For  $x = -\log t/\lambda$ ,  $\lambda \neq 0$ , suppose the positive inner inverse satisfies  $h = A + E$  with  $A > 0$  and  $E/A \rightarrow 0$ . Then*

$$x = -\frac{\log A}{\lambda} - \frac{1}{\lambda} \log \left( 1 + \frac{E}{A} \right) = -\frac{\log A}{\lambda} + \mathcal{O}(E/A). \quad (15.5)$$

*Thus the inner remainder must be divided by the inner leading scale before it becomes an error in  $x$ .*

*Proof.* Factor  $h = A(1 + E/A)$  and expand the logarithm of the positive unit factor. The hypothesis  $E/A \rightarrow 0$  also guarantees  $h > 0$  eventually.  $\square$

### 15.4 Pure logarithmic functions

For  $x \rightarrow +\infty$ , set  $t = \log x \rightarrow +\infty$ . For a positive source distance  $u \rightarrow 0^+$ , set  $t = -\log u \rightarrow +\infty$ , then reconstruct  $u = e^{-t}$  and the original point  $x = x_0 \pm u$ . A pure logarithmic forward expression becomes a power-log expression in  $t$ . Its inverse is computed there before exponential reconstruction.

**Example 15.9** (The constant term in an exponent matters). For  $F(x) = (\log x)^2 + \log x$  at  $+\infty$ , the transformed equation is  $t^2 + t = y$ . Hence

$$t = \sqrt{y} - \frac{1}{2} + \frac{1}{8\sqrt{y}} + \mathcal{O}(y^{-3/2}),$$

and one valid reconstruction is

$$x = \exp\left(\sqrt{y} - \frac{1}{2} + \frac{1}{8\sqrt{y}}\right) (1 + \mathcal{O}(y^{-3/2})).$$

In contrast,  $t = \sqrt{y} + \mathcal{O}(1)$  does not justify  $x = e^{\sqrt{y}}(1 + o(1))$ : the omitted constant would change the leading multiplicative factor by  $e^{-1/2}$ . An implementation must retain enough inner terms to make its omitted additive error tend to zero, even when the initial term request would otherwise stop earlier.

This precision requirement belongs to reconstruction, not to the mere existence of the inner asymptotic expansion. When a user requests a particular relative order in the reconstructed expression, the implementation must translate that request into an inner additive precision and record both meanings.

## 15.5 Finite sums of exponentials

Let

$$F(x) = \sum_{j=1}^m a_j e^{\mu_j x}, \quad a_j, \mu_j \in \mathbb{R},$$

with equal rates merged. At  $x \rightarrow +\infty$  choose  $\lambda > 0$ , and at  $x \rightarrow -\infty$  choose  $\lambda < 0$ , so  $t = e^{-\lambda x} \rightarrow 0^+$  in either case. Then

$$F\left(-\frac{\log t}{\lambda}\right) = \sum_j a_j t^{-\mu_j/\lambda}. \quad (15.6)$$

This is a finite real-power germ. Rational relations among the rates are unnecessary: local finiteness of the positive correction gaps supplies finite exponent truncations. Its leading coefficient and exponent determine the target side, while the chosen sign of  $\lambda$  preserves the source endpoint. After inversion, Corollary 15.8 transports the error back to  $x$ .

**Example 15.10** (An exact oracle). For  $F(x) = e^x + e^{2x}$  at  $+\infty$ , take  $t = e^{-x}$ . Then  $F = t^{-1} + t^{-2}$ , and an exact inverse is

$$x = \log\left(\frac{\sqrt{1+4y}-1}{2}\right) = \frac{1}{2} \log y - \frac{1}{2\sqrt{y}} + \frac{1}{48y^{3/2}} + \mathcal{O}(y^{-5/2}).$$

The inverse of  $e^x + e^{\sqrt{2}x}$  uses the same coordinate with irrational exponents. Constant terms, duplicate rates and cancellations are handled before extracting the leading transformed monomial.

## 15.6 Result contracts and verification

The transformed result retains its inner inverse, coordinate substitution, original function and selected domains. Its scale is identified as **"Transformed"**. The logarithmic target route has coordinate kind **"TargetLog"**; it uses (15.2). The target expression is recorded explicitly. The stored cutoff belongs to the underlying series coordinate after substitution. An ordinary

`SeriesData` in the original target variable is unavailable when that representation would hide the coordinate change. The package README lists the corresponding property names.

The transformed residual describes the model actually composed, and its metadata states that scope. Numerical checks solve the transformed equation at high precision and verify the original source side; they remain numerical evidence. A separate certificate can use Proposition 15.2 with rigorous derivative and residual bounds. An additive input error for  $F$  must be transported through the target map before it can be treated as an error in  $\Phi$ ; until that contract is supplied, rejecting an unsupported declaration is mathematically necessary.

These coordinate theorems justify the stated reductions independently of how many recognizers are implemented. The staged implementation plan tracks the accepted families, source reconstruction operations, tests and remaining extensions; the existence of a reduction alone does not establish that every corresponding expression has already passed the package’s parser and native test suite.

## 16 Incremental and grouped inverse computation

Three changes reduce repeated work without changing the coefficient theorem or the meaning of a complete power-log block. They concern the ordinary engine with finitely many positive exact gaps  $\delta_i$ . Symbolic depth truncation continues to use the individual multi-index formula.

### 16.1 A persistent exact-weight frontier

Let  $I$  be the set of retained multi-indices, initially empty, and let its candidate frontier initially consist of the zero multi-index. At each step, select the smallest candidate weight  $\sigma$ , promote *every* candidate of that weight into  $I$ , and add all their one-step successors  $\mathbf{k} + e_i$  to the candidate set. Candidates are deduplicated by their entire integer vector; their weights are canonicalized independently.

**Proposition 16.1** (Complete layers and coefficient reuse). *After the layer at weight  $\sigma$  is processed,  $I$  is exactly  $\{\mathbf{k} \in \mathbb{N}^m : \mathbf{k} \cdot \boldsymbol{\delta} \leq \sigma\}$ , and the candidate set is its one-step outer boundary. Each retained multi-index has its coefficient evaluated exactly once. Summing the coefficients of an entire layer before counting a block preserves all resonances and exact cancellations.*

*Proof.* Every nonzero multi-index has a predecessor of strictly smaller weight because all gaps are positive. Inductively, every index of the next candidate weight has already been generated by a processed predecessor. A successor of an index in the current layer has strictly larger weight, so processing one member of the layer cannot expose a previously unseen member of the same layer. Deduplication prevents a second insertion through another predecessor. The coefficient of each weight is the complete finite sum in Theorem 6.1, whether or not that sum is zero.  $\square$

The implementation returns its state by value. It stores the normalized model, assumptions, observable power, retained indices, candidate frontier, and merged blocks. Reusing a state therefore reuses coefficients only for the same model and assumptions. It does not use a global cache that might accidentally mix branches or parameter hypotheses. The resource budget counts distinct retained and candidate indices, as in the original bounded enumeration. Raising a term goal no longer recomputes coefficients of every earlier layer. A cancellation may still require arbitrarily many layers before another nonzero block appears; incremental enumeration does not provide a new termination theorem.



## 16.2 Newton iteration with growing precision

Write  $E(U) = 0$  for (6.11), and let  $\delta_* = \min_i \delta_i$ . The zero jet is correct below  $\delta_*$ . If  $U_s$  is correct strictly below weight  $s$ , one Newton step performed strictly below

$$h = \min(2s, H), \quad U_h = \left[ U_s - \frac{E(U_s)}{E'(U_s)} \right]_{<h},$$

is correct strictly below  $h$ . Indeed,  $E'(U_s)$  has nonzero constant coefficient  $p$ , and the Newton error is quadratic in the old error. Products, unit powers, and logarithmic composition do not decrease the weight of the error. The argument uses exponent valuation, with polynomial logarithms as coefficients; logarithmic degree need not stay bounded as  $h$  increases.

The working cutoffs are  $\min(2\delta_*, H), \min(4\delta_*, H), \dots, H$ , rather than  $H$  at every step. Their construction uses exact comparisons, with no floating-point estimate of the required iteration count. The implementation checks  $[E(U_h)]_{<h} = 0$  after each step before recording the new precision. This is an exact formal consistency check. It is not a numerical error certificate for a particular nonzero value of  $z$ .

A reusable Newton state contains its unit jet and certified formal precision. Refinement to a larger cutoff starts from that jet; a request for a smaller cutoff leaves the stronger state intact, and a caller may truncate a copy. The monomial model has the exact unit jet zero and needs no Newton steps. Other models require a finite cutoff.

## 16.3 Combine resonances before applying Euler operators

Put

$$H_1(z, L) = \sum_i z^{\delta_i} B_i(L), \quad H_1(z, L)^n = \sum_{\sigma} z^{\sigma} A_{n,\sigma}(L).$$

The multinomial theorem gives

$$A_{n,\sigma} = \sum_{\substack{|k|=n \\ k \cdot \delta = \sigma}} \frac{n!}{k!} B^k.$$

At fixed depth  $n$  and weight  $\sigma$ , all summands of (6.1) have the same differential operator. By linearity, the unit bracket of  $g^r$  can therefore be computed as

$$\frac{g(y)^r}{z^r} = 1 + \sum_{n \geq 1} \sum_{\sigma} z^{\sigma} \frac{(-1)^n r}{p^n n!} \prod_{j=1}^{n-1} (\mathcal{D} + r + \sigma + pj) A_{n,\sigma}(L). \quad (16.1)$$

For cutoff  $H$ , successive sparse products compute  $[H_1^n]_{<H}$ . Only  $n$  with  $n\delta_* < H$  can contribute, so the computation is finite. Coefficients of equal weight are combined during each sparse product, before differentiation. Contributions at the same weight but different depths are then combined in the result; they too can cancel.

Equation (16.1) is especially useful when the generators have many additive relations. It does not require commensurable gaps, and it does not replace the public individual-coefficient formula: that formula retains the distinct perturbation markers and their provenance. Grouping can reduce coefficient work without eliminating the separate boundary enumeration used for remainder reporting. It is not a claim of uniform speedup on every input; polynomial coefficient growth and sparse multiplication remain material costs.

The independent engine is selected with `Method -> "GroupedLagrange"`. The default `"Lagrange"` method keeps the individual formula, so their agreement remains a direct check of the regrouping. Both use complete exact weights for term goals; grouping does not change which blocks are requested.

## 16.4 Validation obligations

The regression suite compares every successive incremental region with the independent bounded multi-index enumerator, checks exact algebraic collisions and complete cancellations, and verifies the unique-index resource budget. It compares growing-precision Newton with the individual Lagrange formula for incommensurable logarithmic gaps, including a negative leading power, and checks that refinement preserves every previously certified coefficient. Grouped coefficients are compared with individual coefficients for several observable powers and with Newton independently. These are formal algebra checks; runtime measurements and analytic error bounds are separate evidence.

## 16.5 Recorded native measurements

The reproducible script `Tests/BenchmarkIncremental.wl` compares frozen reference control flow with the new helpers and checks exact result equality. For the public grouped comparison it checks the remainder descriptor as well. One fresh-kernel run on Wolfram 15.0.1 for Windows recorded:

| Fixture                                 | Reference seconds | Updated seconds |
|-----------------------------------------|-------------------|-----------------|
| Eight logarithmic blocks, term goal     | 0.19092           | 0.05692         |
| Newton through twelve weights           | 6.62615           | 3.35497         |
| Seven-power model, target cutoff twelve | 0.35947           | 0.48142         |

The first fixture reduces coefficient evaluations from 95 to 31. All three results agree exactly. The grouped public method is slower on this small fixture, where normalization, boundary enumeration and sparse products remain part of the cost. It stays an explicit alternative; these observations do not justify replacing the default method or asserting a uniform speedup.

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